Scientific Analysis of Moral Judgments using Tobit and Cluster-Aware Non-Parametric Robustness Models

Automated Research Pipeline

marzo 31, 2026

1 Dataset and Sample Description

The empirical foundation of this project rests on two primary experimental datasets: FLORIDA and BUC. The sample consists of students from both the Floridablanca and Bucaramanga Campuses. These datasets capture incentivized moral judgments from distinct socio-economic contexts. Participants were presented with standardized negotiation scenarios where a negotiator's decision resulted in varying degrees of payoff for themselves, their own group, and a victim group. Under Option 2: judgment-level relational modeling , each judgment is treated as relational and linked both to a descriptive victim x negotiator scenario shorthand and to hypothesis-specific predictors for negotiator-side structure, observer-side victim alignment when applicable, and additional relational controls.

2 Datacard and Variable Definitions

The following table defines the primary symbols and variables used in the mathematical specifications and hypothesis tests.

Table 1: Symbols and Variable Dictionary.

Symbol	Definition
y_{isjr}	Observed moral judgment for participant i, scenario s, judged negotiator j, and role r.
y^*_{isjr}	Latent moral preference score (unbounded).
β_1	Regression coefficient representing the marginal effect of the predictor.
$\beta_{0.5}$	Median-regression coefficient from the non-parametric censored robustness model.
$Q_{0.5}(y^*_{isjr} \mid \mathbf{x}_{isjr})$	Conditional median of the latent bounded outcome given the predictors.
IRI_i	Empathy score (Average composite of the Interpersonal Reactivity Index).
CaseConfig_{isjr}	Judgment-level relational shorthand retained only for backward compatibility with older descriptive artifacts.
$\mathbf{S}_{isj}$	Negotiator-side structure block encoding the judged and counterpart negotiators jointly within a judgment row.
V_{is}	Observer-side player-victim outgroup indicator used only in the bystander H2 specification.
G_{isjr}	Judged negotiator relational status (ingroup, outgroup, or control) defined relative to the role-relevant reference actor for H3.
A_{is}	Decision indicator distinguishing Accept from Reject for H3.
C_{isjr}	Additional controls, including participant demographics, slot effects, and the remaining relational terms not absorbed by the main H2/H3 blocks.
ICC	Intraclass Correlation: ratio of between-cluster variance to total variance.

Option 2: judgment-level relational modeling retains a victim x judged-negotiator shorthand for descriptive summaries, while the executable hypotheses use explicit negotiator-level structure terms. H2 now uses the judged-plus-counterpart structure directly, and in observer rows it adds the player-victim alignment term and its interaction with that structure.

2.1 Predictor Glossary and Abbreviations

The regression tables and dynamic figures use compact predictor labels to stay readable. The bullet list below maps those compact labels back to their full meaning and subset-specific interpretation.

- `iri_total` [Emp]: Composite empathy predictor used in Model A. Used in H1/H2/H3 Model A.
- `iri_fs, iri_ec, iri_pt, iri_pd` [FS, EC, PT, PD]: IRI subscales used in Model B: fantasy, empathic concern, perspective taking, and personal distress. Used in H1/H2/H3 Model B.
- `judged_outgroup, judged_control` [JN Out, JN Ctl]: Judged-negotiator contrasts relative to judged ingroup. Used in H1/H3.
- `counterpart_outgroup, counterpart_control` [CN Out, CN Ctl]: Counterpart-negotiator contrasts relative to counterpart ingroup. Used in H1/H3.
- `decision_accept` [Acc]: Accepted harmful deal relative to rejected harmful deal. Used in H1/H3.
- `observer_victim_outgroup` [V Out (Obs)]: Observer-only victim outgroup contrast; excluded from victim-only formulas. Used in H1/H3 Bystander only.
- `h2_negstruct_*` [JN ..., CN ...]: H2 joint judged/counterpart structure dummies with reference JN In, CN In. Used in H2 Victim and Bystander.
- `player_victim_outgroup` [V Out]: Observer-side player-victim outgroup contrast with player-victim ingroup as the reference. Used in H2 Bystander only.
- `player_victim_outgroup:h2_negstruct_*` [V Out x JN ..., CN ...]: Bystander-only H2 interaction: whether the negotiator-side structure changes when player and victim are outgroup-aligned. Used in H2 Bystander only.
- `decision_accept:judged_*` [Acc x JN ...]: H3 decision-by-judged-status interaction block. Used in H3.
- `iri_*:judged_*` [Emp / FS / EC / PT / PD x JN ...]: H3 empathy-by-judged-status interaction block. Used in H3.
- `participant_engineering` [Eng part.]: Participant faculty contrast. Used in H1/H2/H3.
- `sex_man` [Man]: Sex contrast with woman as the reference. Used in H1/H2/H3.
- `age` [Age]: Participant age in original units. Used in H1/H2/H3.
- `economic_status` [SES]: Economic status in original units. Used in H1/H2/H3.
- `factor(negotiator_slot)` [Slot 2]: Negotiator slot contrast with slot 1 as the reference. Used in H1/H2/H3.

2.2 Interpretation of Interaction Terms

The models herein employ several predefined predictors. It is important to note how interaction terms are interpreted in the context of this behavioral experiment:

1. **Interaction Subsumption:** When an interaction term is statistically significant, it indicates that the effect of one variable depends on the level of the other. Crucially, if the interaction is significant but the constituent main effects are not explicitly significant, their effects are fully subsumed and contextualized by the interaction.

2. **Continuous by Discrete Interactions:** For terms like `iri_total:judged_outgroup`, a negative coefficient implies that the severity of moral judgment (lower score) induced by higher empathy is steeper (magnified) when evaluating an outgroup negotiator compared to an ingroup negotiator. A positive coefficient would mean empathy makes judgments less severe for the outgroup.

3. **Discrete by Discrete Interactions:** For terms like `judged_outgroup:decision_accept`, a positive coefficient implies that the change in moral judgment when moving from rejecting a deal to accepting a deal is more positive (less morally condemned) for an outgroup negotiator than for an ingroup negotiator.

3 Hypotheses to Test

- **H1 (Empathy under relational controls):** Within the victim and bystander subsets, higher empathy predicts lower moral-judgment scores for harmful decisions after conditioning on judged-negotiator status, counterpart status, decision outcome, observer-side victim alignment when applicable, and participant controls.

- **H2 (Negotiator-side relational structure):** Moral-judgment severity will differ as a function of the judgment-level ingroup/outgroup/control structure of the judged and counterpart negotiators, with an additional player-victim alignment interaction in the bystander subset.

- **H3 (Empathy x judged-status moderation):** The empathy effect may vary across judged-negotiator relational status, while decision outcome and relational controls remain in the model.

4 Mathematical Approach and Theoretical Foundations

The analysis employs two complementary estimators for bounded moral judgments. The primary specification is a Two-Limit Tobit model, which is theoretically appropriate for dependent variables that are strictly bounded within a known interval. In this context, negotiator-specific moral judgments (y_{isjr}) are observed on a scale from -9 to 9. The Tobit model assumes the existence of a latent, unobserved preference index (y^*_{isjr}) that follows a linear relationship. Option 2 replaces isolated ingroup/outgroup indicators with explicit judgment-level relational variables built from the paired-group structure of each judgment. The analytic hypothesis models decompose that structure into judged-negotiator status, counterpart-negotiator status, observer-side victim alignment when applicable, and hypothesis-specific interaction blocks. H1, H2, and H3 are all estimated separately in the victim and bystander subsets. H1 and H3 use subset-specific formulas so observer-only victim-alignment predictors are retained only where they vary, while H2 uses explicit subset-specific relational-structure blocks:

Victim-subset H2:

$$y^*_{isj,Victim} = \beta_0 + \beta_1 \text{Empathy}_i + \boldsymbol{\gamma}'\mathbf{S}^{(V)}_{isj} + \boldsymbol{\delta}'\mathbf{Z}_i + \epsilon_{isj}$$

Bystander-subset H2:

$$y^*_{isj,Obs} = \beta_0 + \beta_1 \text{Empathy}_i + \boldsymbol{\gamma}'\mathbf{S}^{(O)}_{isj} + \eta V_{is} + \boldsymbol{\theta}'(\mathbf{S}^{(O)}_{isj} \times V_{is}) + \boldsymbol{\delta}'\mathbf{Z}_i + \epsilon_{isj}$$

H3:

$$y^*_{isjr} = \beta_0 + \beta_1 \text{Empathy}_i + \boldsymbol{\beta}'_2 G_{isjr} + \beta_3 A_{is} + \boldsymbol{\beta}'_4 (G_{isjr} \times A_{is}) + \boldsymbol{\beta}'_5 (\text{Empathy}_i \times G_{isjr}) + \boldsymbol{\beta}'_6 C_{isjr} + \epsilon_{isjr}, \quad \epsilon_{isjr} \sim N(0, \sigma^2)$$

The actual observed judgment y_{isjr} relates to this latent variable via the censoring transformation:

$$y_{isjr} = \max(-9, \min(9, y^*_{isjr}))$$

This approach prevents the 'ceiling' and 'floor' effects from biasing the linear coefficients, as would occur in standard OLS regression.

Because the Tobit model relies on a Gaussian latent-error assumption, the pipeline also fits a distribution-robust censored median specification implemented as interval-censored quantile regression ($p = 0.5$). This complementary estimator targets the conditional median of the latent bounded outcome and is less sensitive to heavy tails and non-normal disturbances:

$$Q_{0.5}(y^*_{isjr} \mid \mathbf{x}_{isjr}) = \mathbf{x}'_{isjr}\beta_{0.5}$$

The non-parametric branch preserves the censoring structure while relaxing the parametric normality assumption. The Tobit and robustness results should therefore be interpreted jointly: Tobit provides the clustered parametric benchmark with participant-clustered standard errors by id, while the non-parametric branch first establishes a converged full-sample censored median fit and then, in the default pipeline, adds participant-level cluster bootstrap inference that resamples ids with replacement and retains all repeated observations from each sampled participant. In both branches, id is used only to account for within-participant dependence and is not a substantive explanatory variable.

5 Analysis of Sample Size Impact

The statistical validity of our inferences depends on the trade-off between False Positives (Type I error, α) and False Negatives (Type II error, β). Given the clustered nature of our data, sample size impact is not merely a count of observations, but a function of cluster correlation.

5.0.1 Step-by-Step Sensitivity Analysis

To determine the impact of our sample size on our ability to detect effects, we follow these steps:

1. Calculate the Design Effect (Deff): As multiple judgments are nested within individuals, we adjust for the Intraclass Correlation (ICC).

$$Deff = 1 + (m - 1) \times ICC$$

where m is the average number of scenarios per participant.

2. Determine the Effective Sample Size (ESS): The ESS represents the number of independent observations that would provide the same statistical power as our clustered sample.

$$ESS = \frac{n_{total}}{Deff}$$

3. Inference Impact: A higher ICC reduces the ESS, thereby increasing the Standard Error of our Tobit coefficients. If the ESS is low, our models become 'conservative', increasing the risk of Type II errors (failing to support a hypothesis). By using clustered robust standard errors for Tobit and participant-level cluster bootstrap inference for the non-parametric robustness model after each converged full-sample fit, we ensure that both sets of p-values acknowledge this reduced information density, protecting the integrity of our Type I error threshold ($\alpha = 0.05$).

Based on the current run, the observed Intraclass Correlation (ICC) is 0.090, with an Effective Sample Size (ESS) of 1891.5.

6 Descriptive Statistics

The empathy profile of the sample is visualized in Figure 1, showing the average scores across the four IRI latent variables.

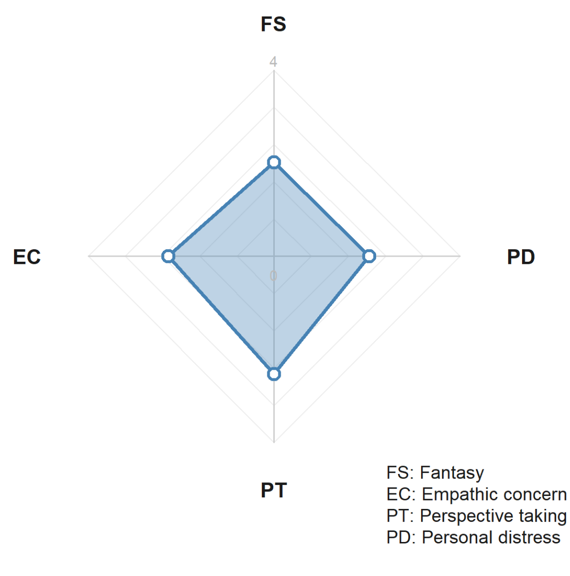
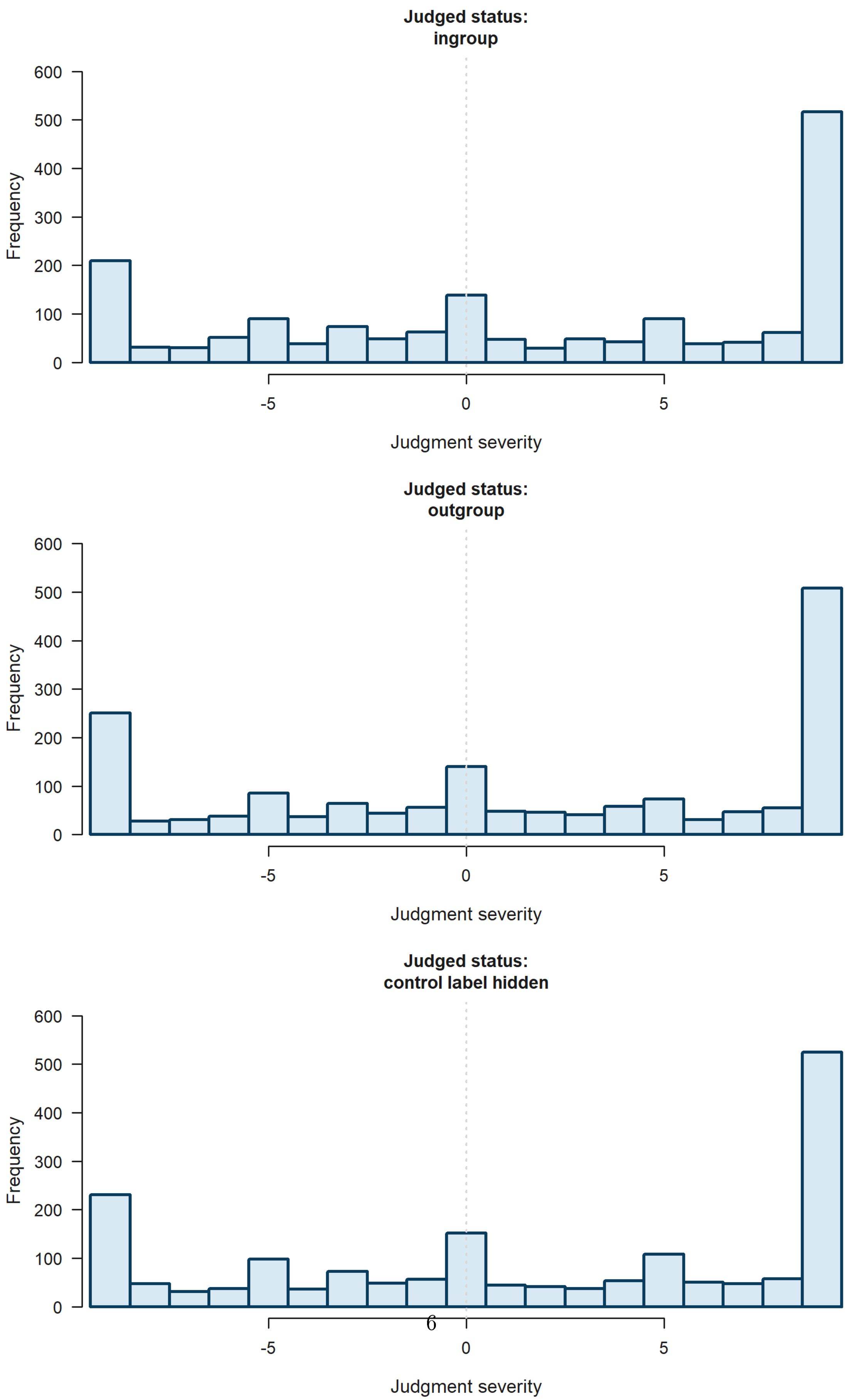


Figure 1: IRI Latent Variable Averages (Radar Plot profile).

Figure 2 summarizes the overall judgment distribution across judged-negotiator status. Figures 3 and 4 replicate that severity-panel logic across the six explicit victim x negotiator case configurations within the victim and bystander subsets, and Figures 5 and 6 add the same six-panel histograms for accepted versus rejected harmful deals.


Judged status:
ingroup
Frequency
600
500
400
300
200
100
0
-5
0
5
Judgment severity
Judged status:
outgroup
Frequency
600
500
400
300
200
100
0
-5
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Judgment severity
Judged status:
control label hidden
Frequency
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Judgment severity

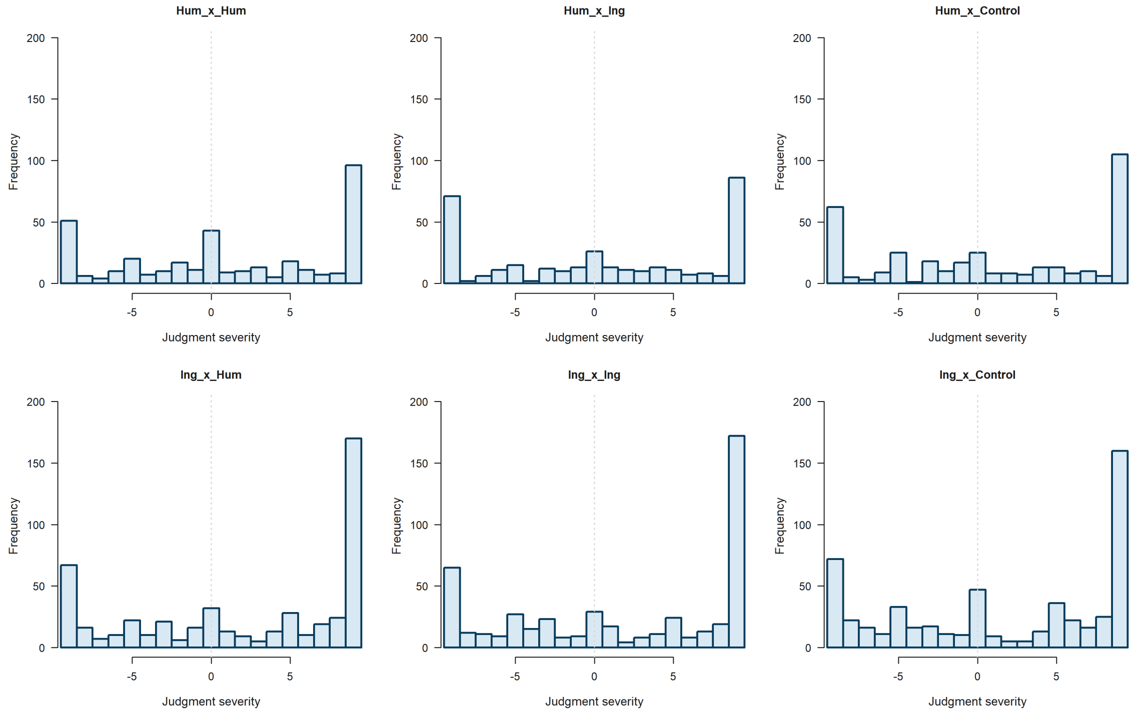

Figure 3: Victim-subset severity panels across the six explicit victim x negotiator case configurations, using a fixed judgment scale from -9 to 9.

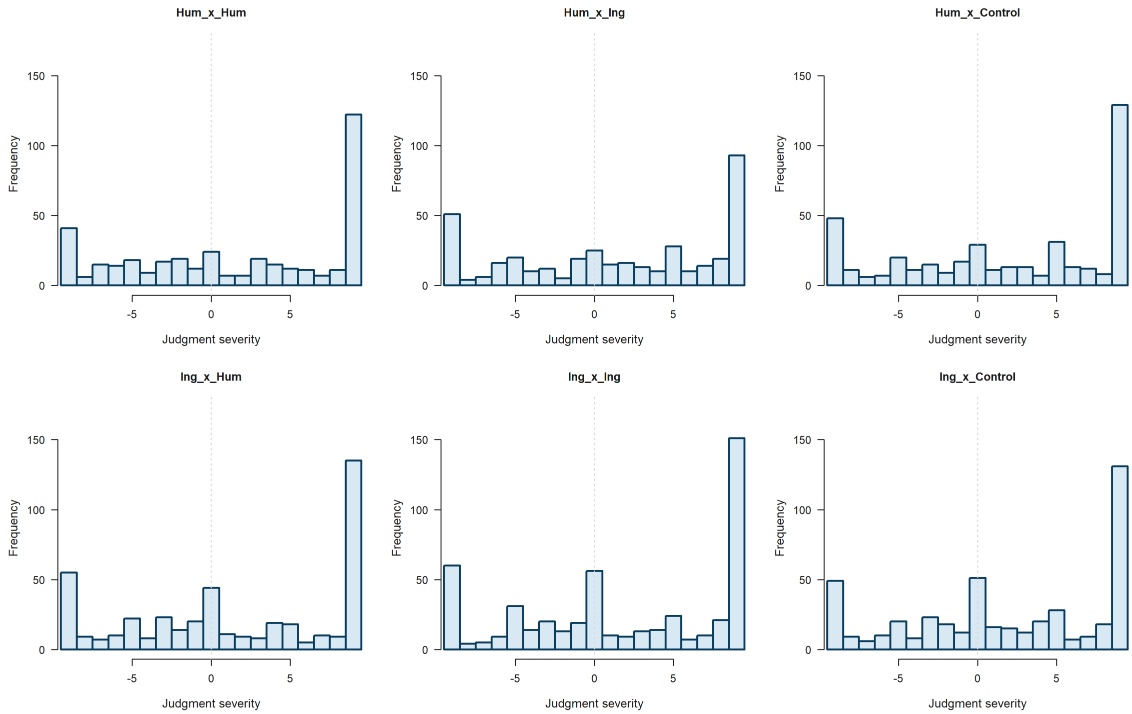

Figure 4: Bystander-subset severity panels across the six explicit victim x negotiator case configurations, using a fixed judgment scale from -9 to 9.

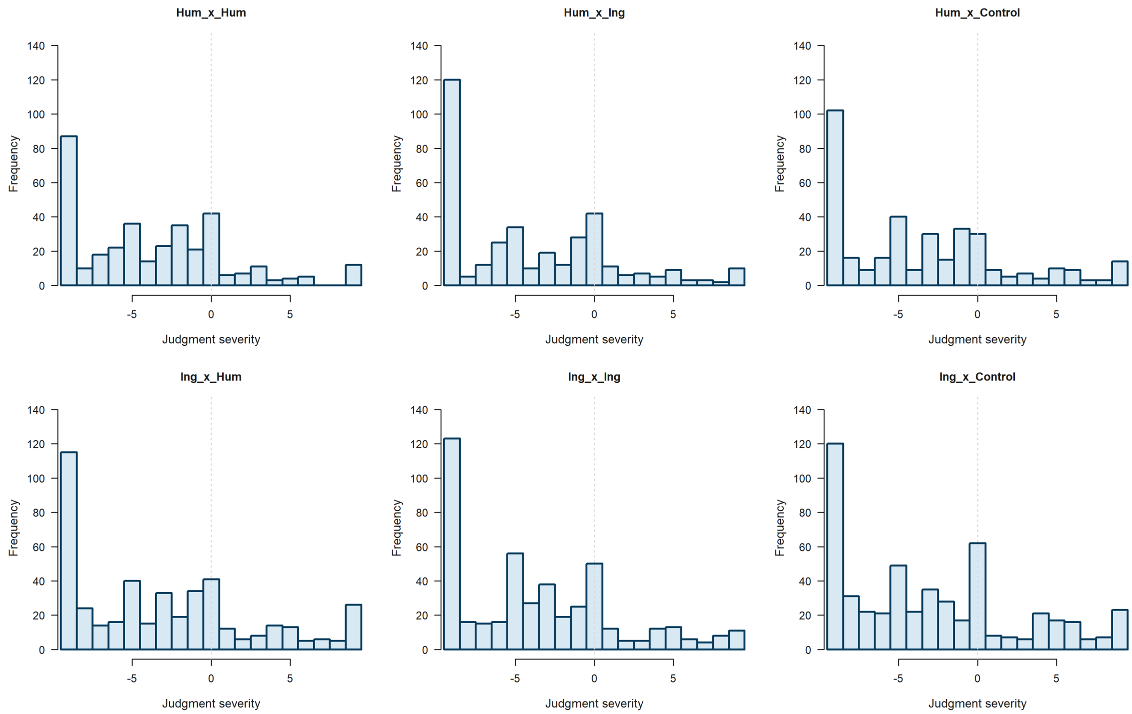

Figure 5: Agreement severity panels across the six explicit victim x negotiator case configurations, using a fixed judgment scale from -9 to 9.

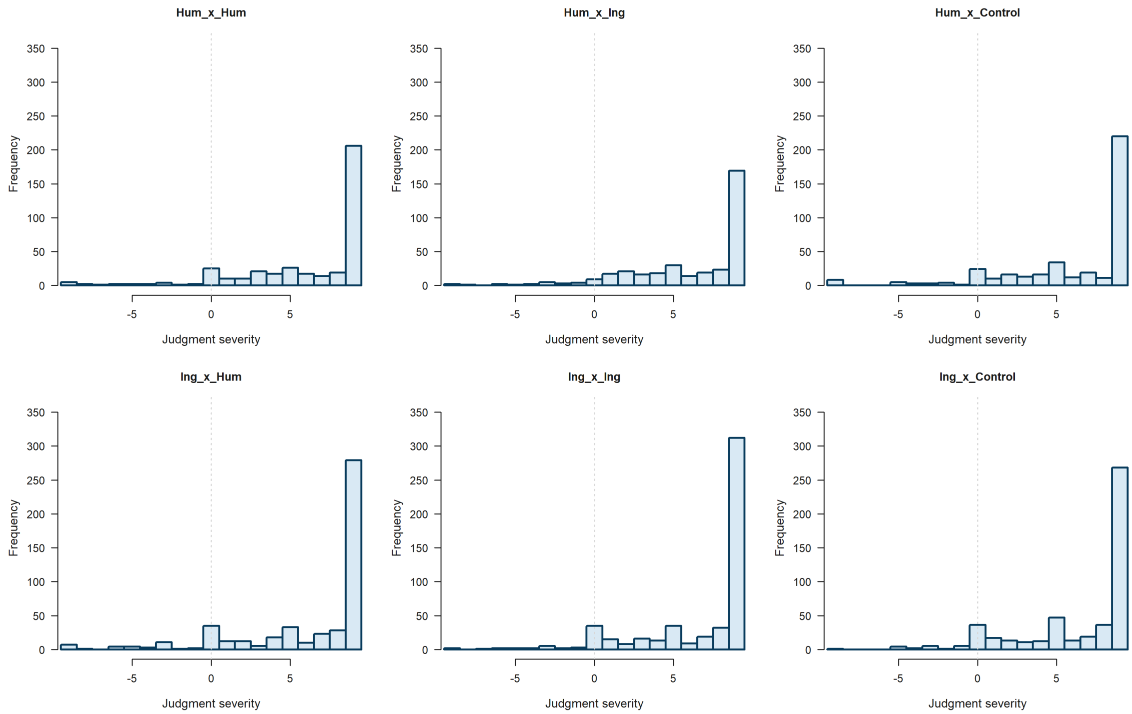

Figure 6: Disagreement severity panels across the six explicit victim x negotiator case configurations, using a fixed judgment scale from -9 to 9.

The descriptive branch no longer centers the main report on victim x negotiator case-label tables; the hypothesis tests below work directly with negotiator-level relational predictors.

7 Bi-variate Statistics

The correlation matrix between the psychometric subscales and the mean moral judgment is presented below.

Table 2: Correlation Matrix: IRI Subscales and Moral Judgment.

Fantasy	Empathic Concern	Perspective Taking	Personal Distress	Total IRI	Mean Judgment
1.000	0.455	0.541	0.468	0.821	-0.041
0.455	1.000	0.387	0.614	0.792	-0.191
0.541	0.387	1.000	0.257	0.721	0.103
0.468	0.614	0.257	1.000	0.736	0.032
0.821	0.792	0.721	0.736	1.000	-0.035
-0.041	-0.191	0.103	0.032	-0.035	1.000

Visual representations of these relationships, including fitted lines with 95% confidence bands, are provided in Figure 7.

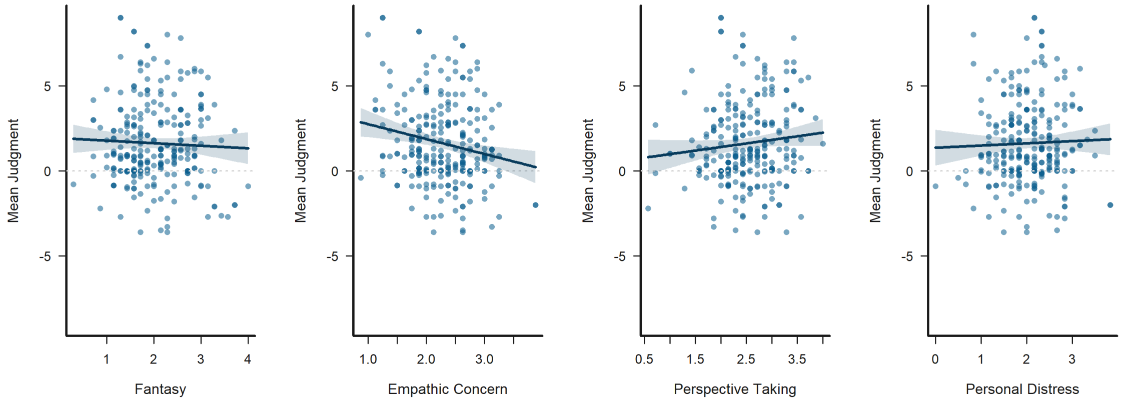

Figure 7: Bivariate Scatters: IRI Scales vs. Mean Judgment, with fitted lines and 95\{}% confidence bands.

8 Estimator Fit Summary

The following table consolidates the fit-status information for the primary Tobit models used in this report.

Table 3: Estimator fit summary across Tobit specifications.

Model	Approach	Status	Iterations	Observations	Participants	ClusterUnit
H1_A_Bystander_Total_Tobit	Tobit	completed	5	2570	257	id
H1_A_Victim_Total_Tobit	Tobit	completed	6	2570	257	id
H1_B_Bystander_Constructs_Tobit	Tobit	completed	5	2570	257	id
H1_B_Victim_Constructs_Tobit	Tobit	completed	6	2570	257	id
H2_A_Bystander_Total_Tobit	Tobit	completed	3	2570	257	id
H2_A_Victim_Total_Tobit	Tobit	completed	3	2570	257	id
H2_B_Bystander_Constructs_Tobit	Tobit	completed	3	2570	257	id
H2_B_Victim_Constructs_Tobit	Tobit	completed	3	2570	257	id
H3_A_Bystander_Total_Tobit	Tobit	completed	5	2570	257	id
H3_A_Victim_Total_Tobit	Tobit	completed	6	2570	257	id
H3_B_Bystander_Constructs_Tobit	Tobit	completed	5	2570	257	id
H3_B_Victim_Constructs_Tobit	Tobit	completed	6	2570	257	id

8.1 Hypothesis Significance Summary

The following concise tables list only hypothesis-relevant predictors that reached at least $p < 0.10$ in the available Tobit models, split by victim and bystander subsets, using conventional symbols to indicate strength of evidence (+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Dynamic figures are generated only for predictors that appear in these tables with at least one significance symbol.

Table 4: Hypothesis-level significance summary for the victim subset across Tobit models.

Hypothesis	Tobit support
H1	PT***; EC***; PD**
H2	JN Ctl, CN In+
H3	Emp x JN Out*; Emp x JN Ctl*; EC x JN Out+

Victim subset

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Table 5: Hypothesis-level significance summary for the bystander subset across Tobit models.

Hypothesis	Tobit support
H1	EC*; PT*; PD*
H2	JN Out, CN In+
H3	Emp x JN Ctl*; PD x JN Out*

Bystander subset

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

8.2 Significance-Driven Figures

Only hypothesis-relevant predictors that reach $p < 0.10$ or better are visualized automatically. These dynamic figures use the saved Tobit outputs, and participant id remains only an inference-level clustering unit rather than a substantive explanatory variable.

H1: Empathy under relational controls Empathy: Perspective taking is statistically significant in the Tobit model (***, $p < 0.001$). Figure 8 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

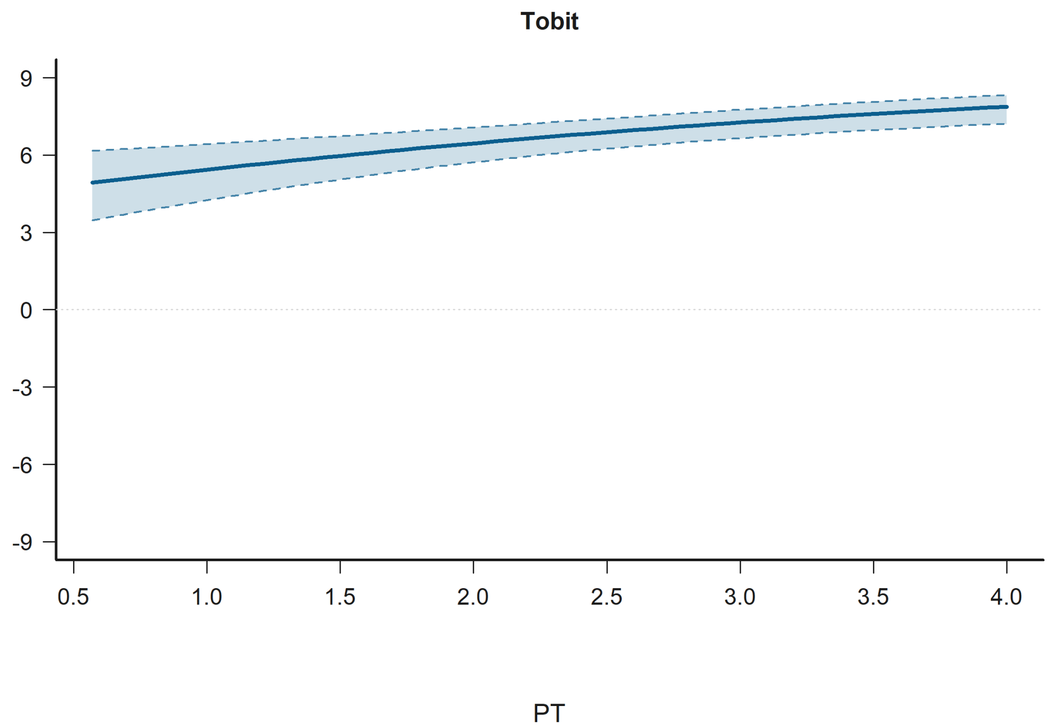

Figure 8: Effect Plot for Empathy: Perspective taking in H1: Empathy under relational controls. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Empathic concern is statistically significant in the Tobit model (***, $p < 0.001$). Figure 9 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted judgment.

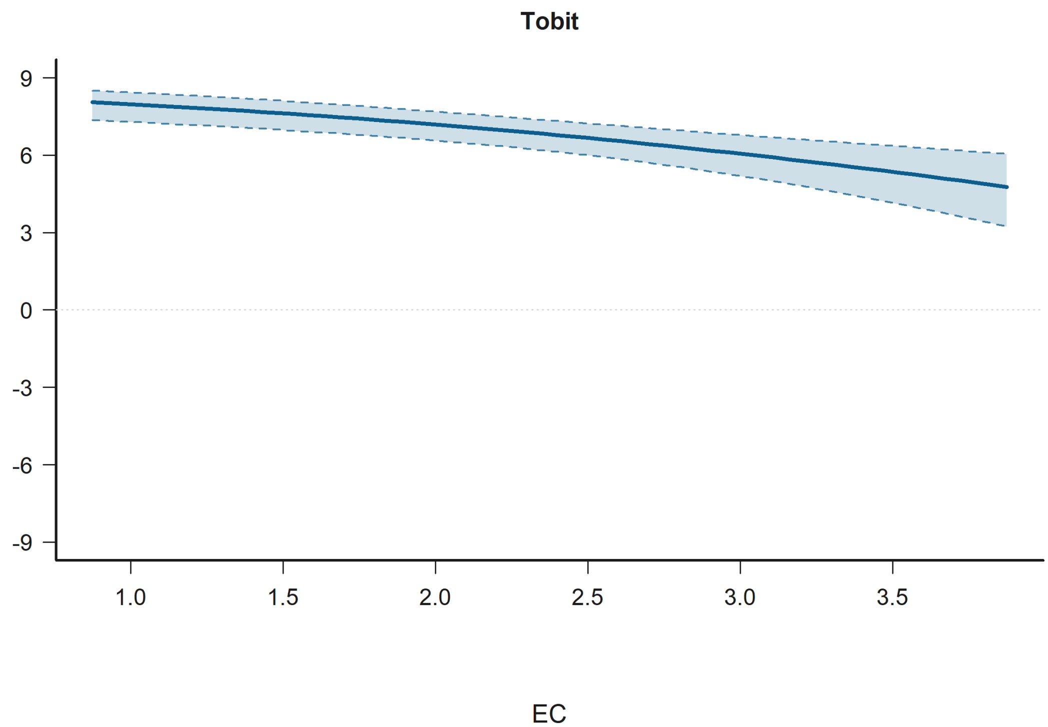

Figure 9: Effect Plot for Empathy: Empathic concern in H1: Empathy under relational controls. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Personal distress is statistically significant in the Tobit model (**, p = 0.004). Figure 10 shows that across the observed range, higher Empathy: Personal distress corresponds to higher predicted judgment.

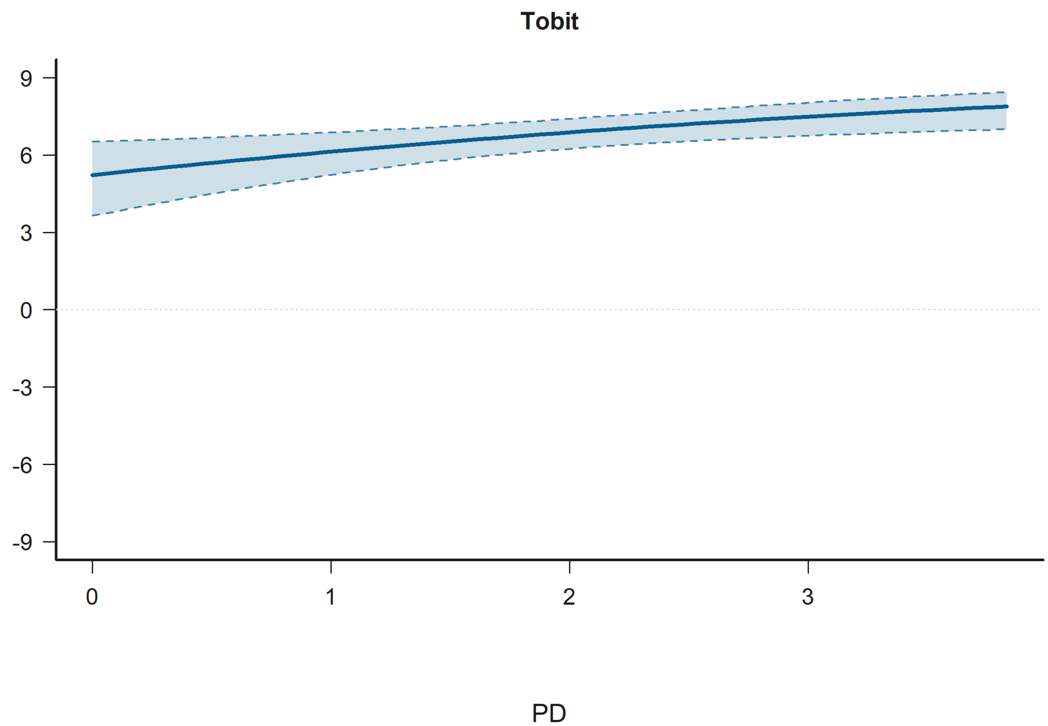

Figure 10: Effect Plot for Empathy: Personal distress in H1: Empathy under relational controls. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

H2: Negotiator-side relational structure Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) is statistically significant in the Tobit model (+, $p = 0.065$). Figure 11 shows that predicted judgment is lower for JN Out, CN In than for Ref: JN In, CN In.

Figure 11: Grouped Prediction Plot for Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) in H2: Negotiator-side relational structure. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) is statistically significant in the Tobit model (+, p = 0.087). Figure 12 shows that predicted judgment is lower for JN Ctl, CN In than for Ref: JN In, CN In.

Figure 12: Grouped Prediction Plot for Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) in H2: Negotiator-side relational structure. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

H3: Empathy x judged-status moderation Empathy x judged negotiator control label hidden is statistically significant in the Tobit model (*, p = 0.015). Figure 13 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is JN Ctl.

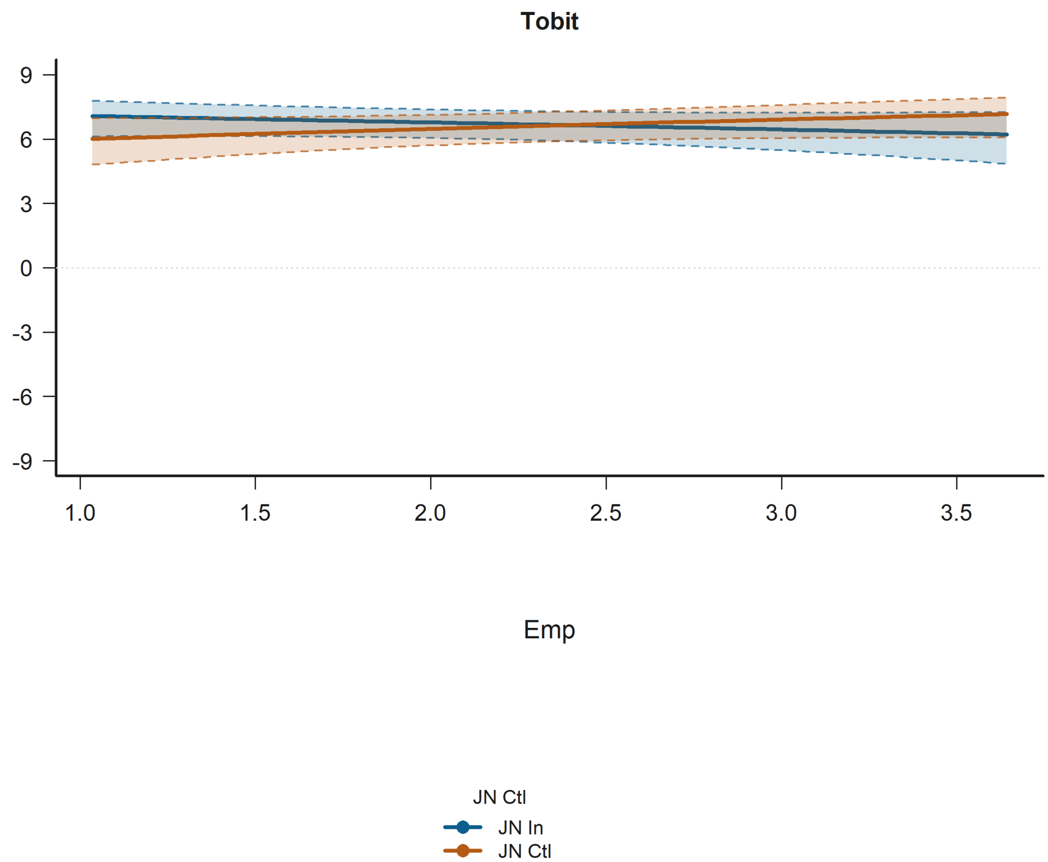

Figure 13: Interaction Plot for Empathy x judged negotiator control label hidden in H3: Empathy x judged-status moderation. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Personal distress x judged negotiator outgroup is statistically significant in the Tobit model (*, p = 0.021). Figure 14 shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is JN Out.

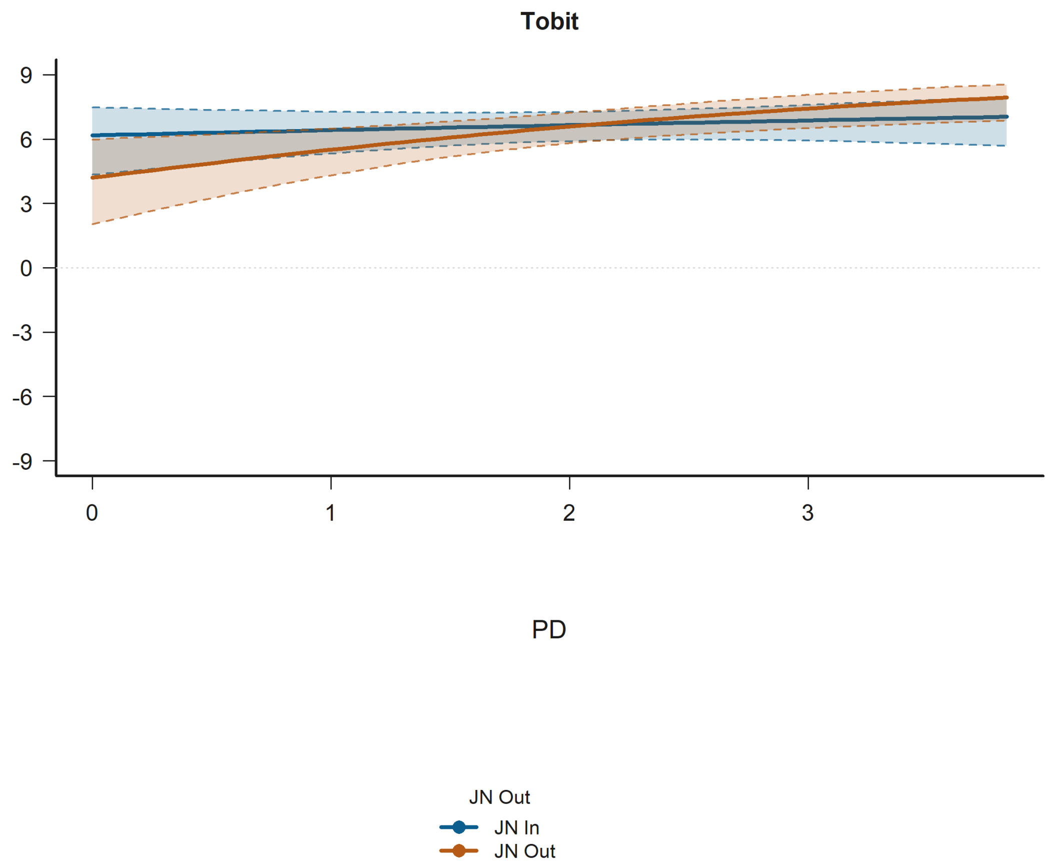

Figure 14: Interaction Plot for Personal distress x judged negotiator outgroup in H3: Empathy x judged-status moderation. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy x judged negotiator outgroup is statistically significant in the Tobit model (*, p = 0.028). Figure 15 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is JN In.

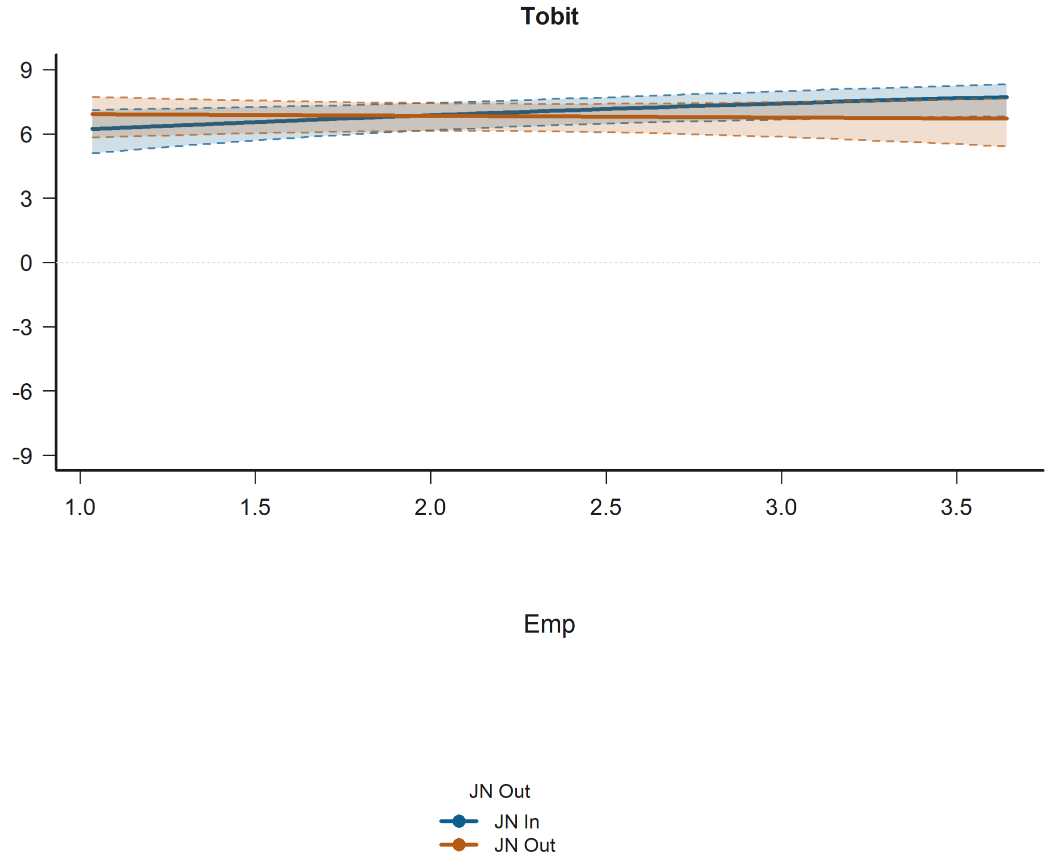

Figure 15: Interaction Plot for Empathy x judged negotiator outgroup in H3: Empathy x judged-status moderation. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathic concern x judged negotiator outgroup is statistically significant in the Tobit model (+, p = 0.053). Figure 16 shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is JN Out.

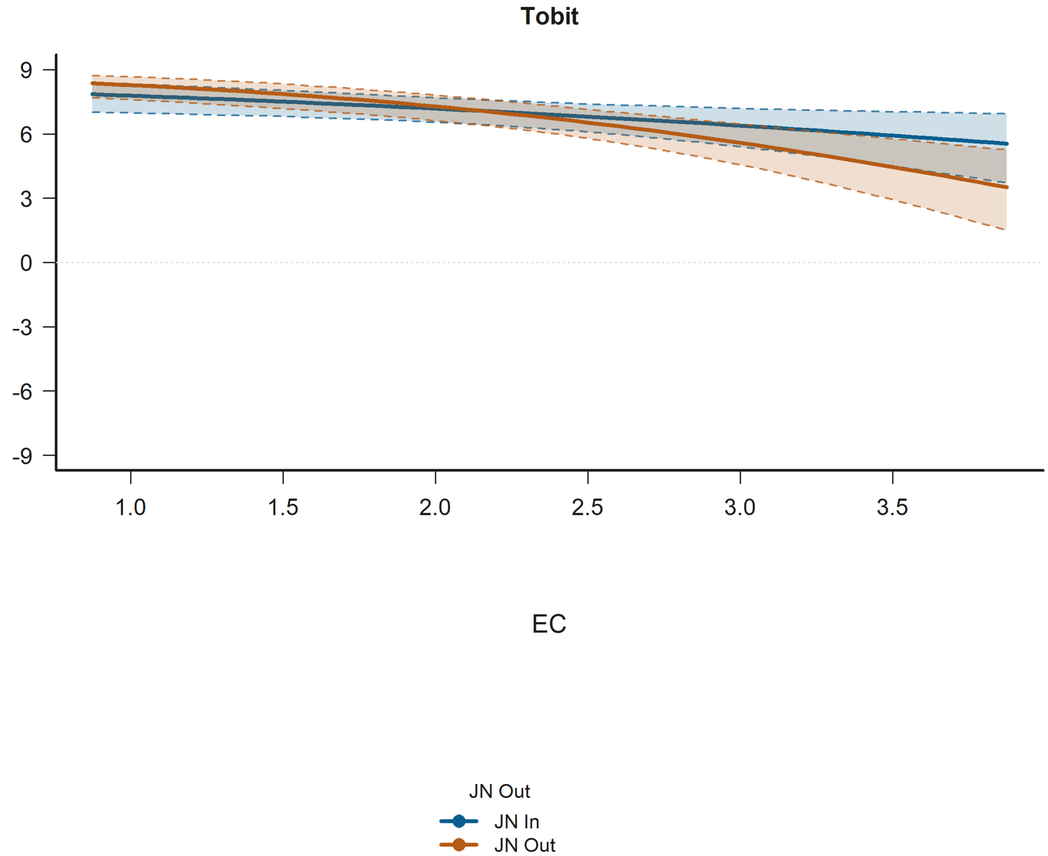

Figure 16: Interaction Plot for Empathic concern x judged negotiator outgroup in H3: Empathy x judged-status moderation. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

8.3 All Significant Predictors ($p < 0.10$)

The following figures extend beyond the hypothesis-target terms and visualize every predictor that reaches $p < 0.10$ in the available H1-H3 Tobit models. This includes significant controls such as age when they clear the threshold.

judgments_bystander Negotiator accepted harmful deal is statistically significant in:

- H1: Empathy under relational controls Model A (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 17 shows that predicted judgment is lower for Acc than for Rej.

Figure 17: Grouped Prediction Plot for Negotiator accepted harmful deal in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Age is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H2: Negotiator-side relational structure Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 18 shows that across the observed range, higher Age corresponds to higher predicted judgment.

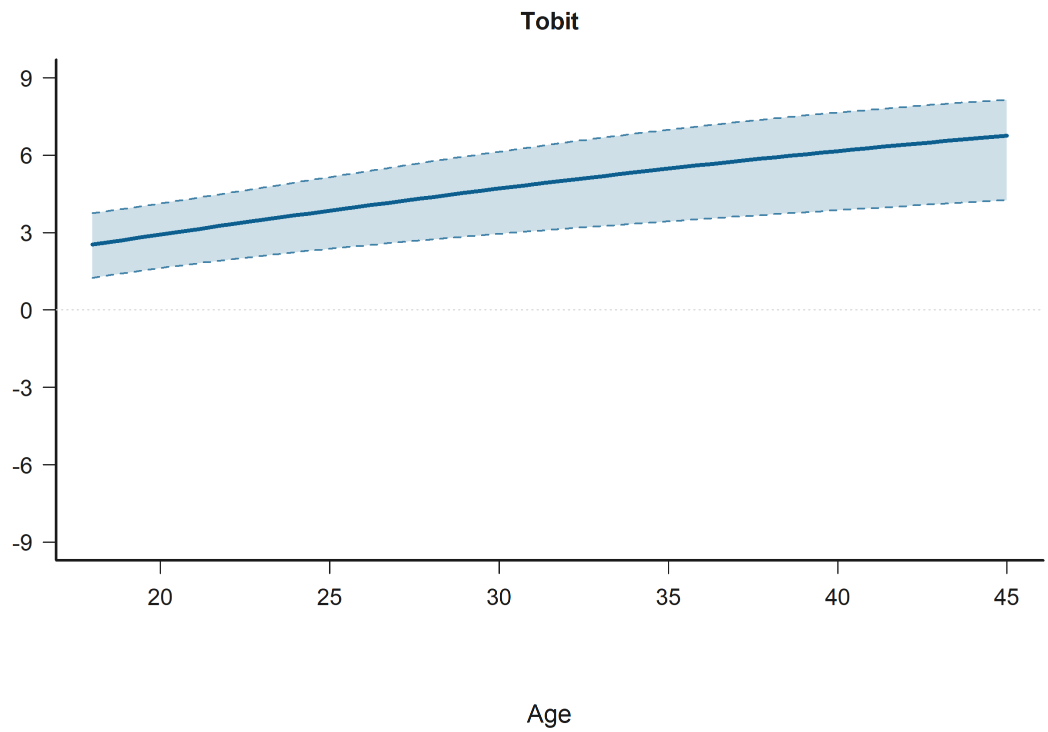

Figure 18: Effect Plot for Age in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Empathic concern is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)

Figure 19 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted judgment.

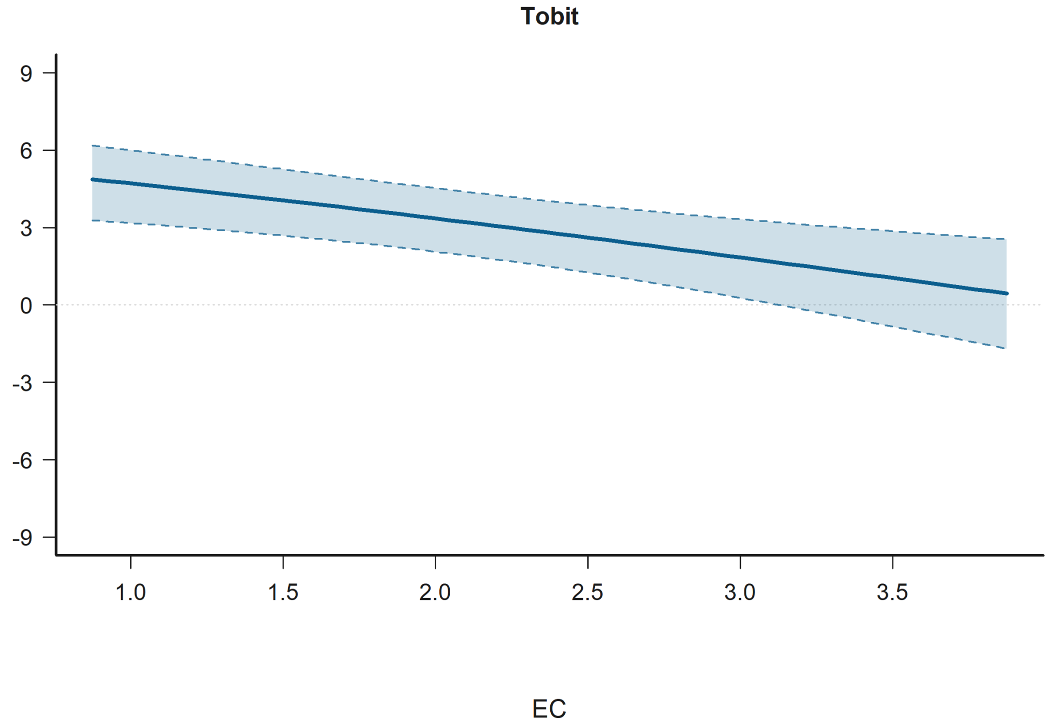

Figure 19: Effect Plot for Empathy: Empathic concern in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Judged negotiator control label hidden (ref = ingroup) is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 20 shows that predicted judgment is lower for JN Ctl than for JN In.

Figure 20: Grouped Prediction Plot for Judged negotiator control label hidden (ref = ingroup) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Personal distress is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)

Figure 21 shows that across the observed range, higher Empathy: Personal distress corresponds to higher predicted judgment.

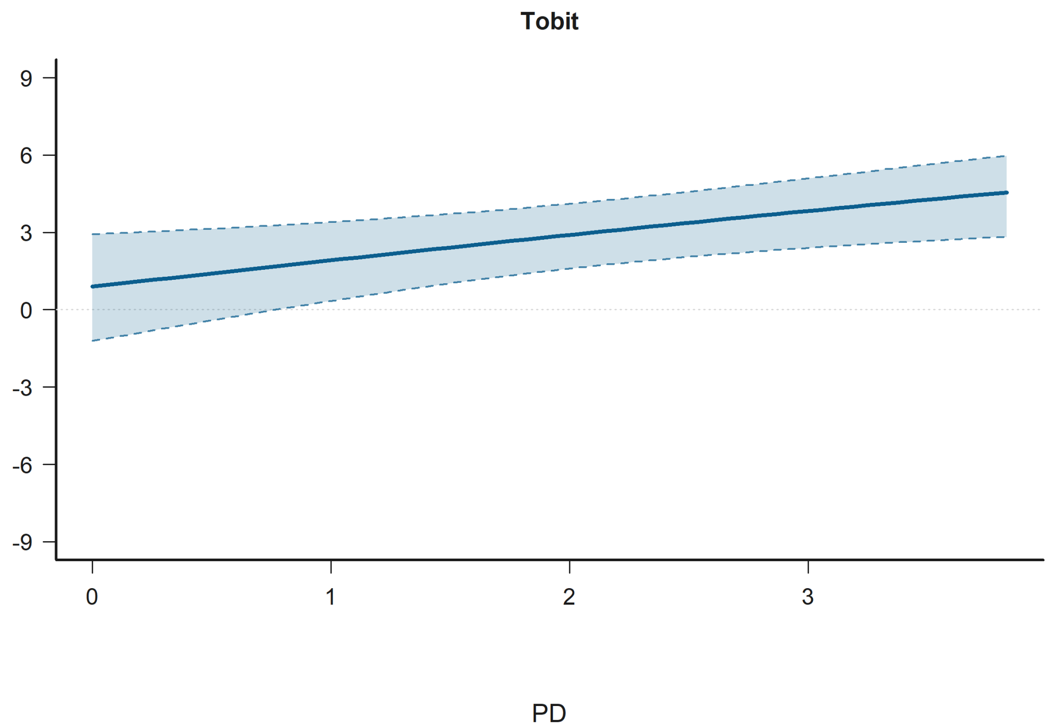

Figure 21: Effect Plot for Empathy: Personal distress in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy x judged negotiator control label hidden is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 22 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is JN Ctl.

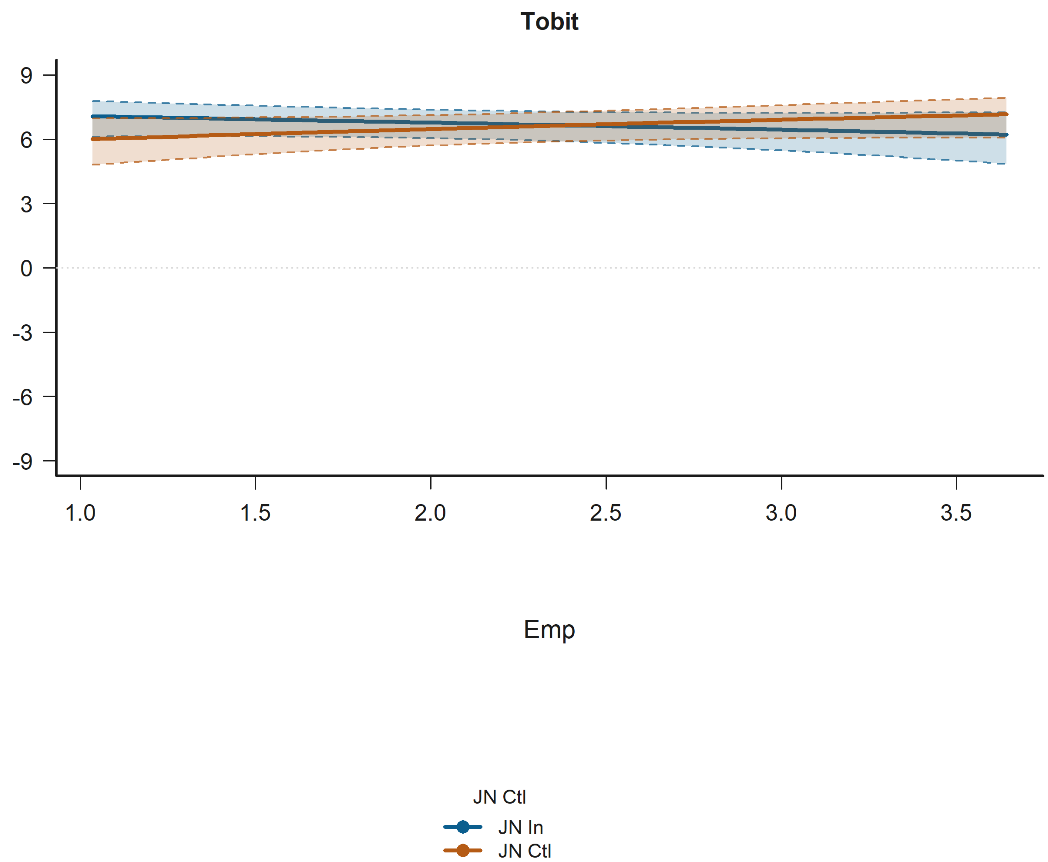

Figure 22: Interaction Plot for Empathy x judged negotiator control label hidden in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Engineering participant (ref = humanities) is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)
- H1: Empathy under relational controls Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H2: Negotiator-side relational structure Model A (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)

Figure 23 shows that predicted judgment is higher for Eng part. than for Hum part..

Figure 23: Grouped Prediction Plot for Engineering participant (ref = humanities) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Personal distress x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 24 shows that the predicted relationship rises most sharply for Empathy: Personal distress when the condition is JN Out.

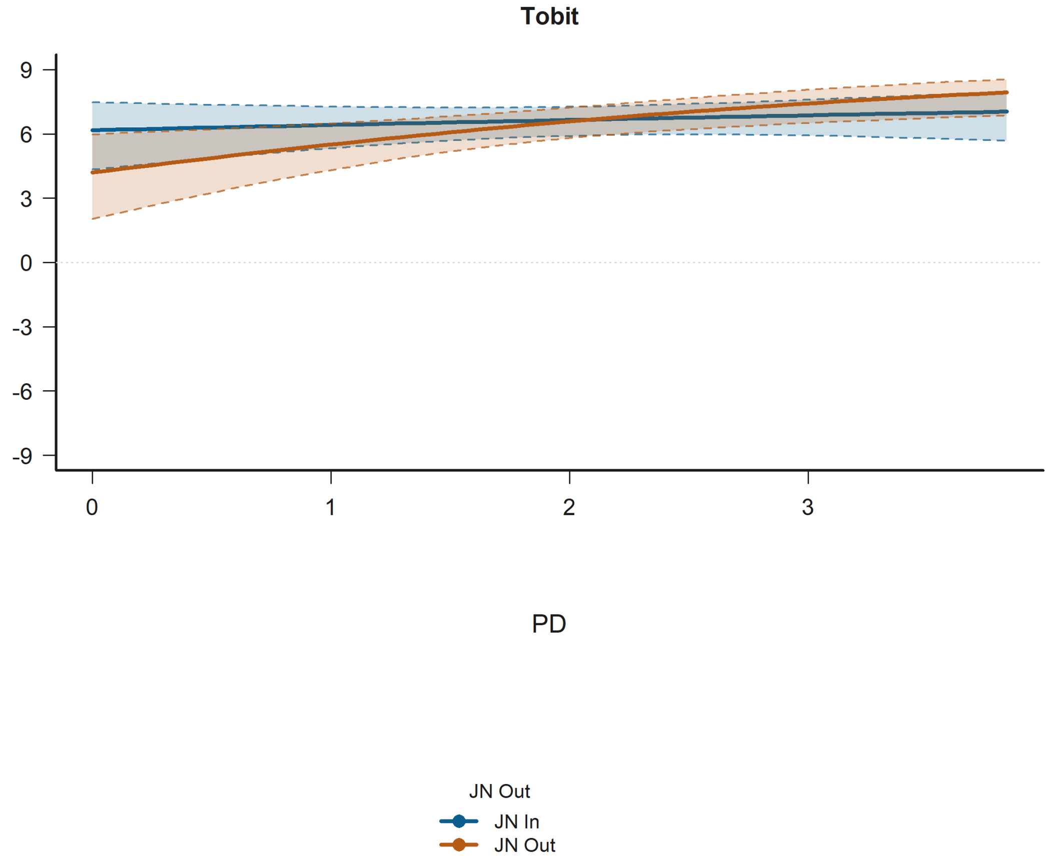

Figure 24: Interaction Plot for Personal distress x judged negotiator outgroup in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Perspective taking is statistically significant in:

- H1: Empathy under relational controls Model B (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)

Figure 25 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

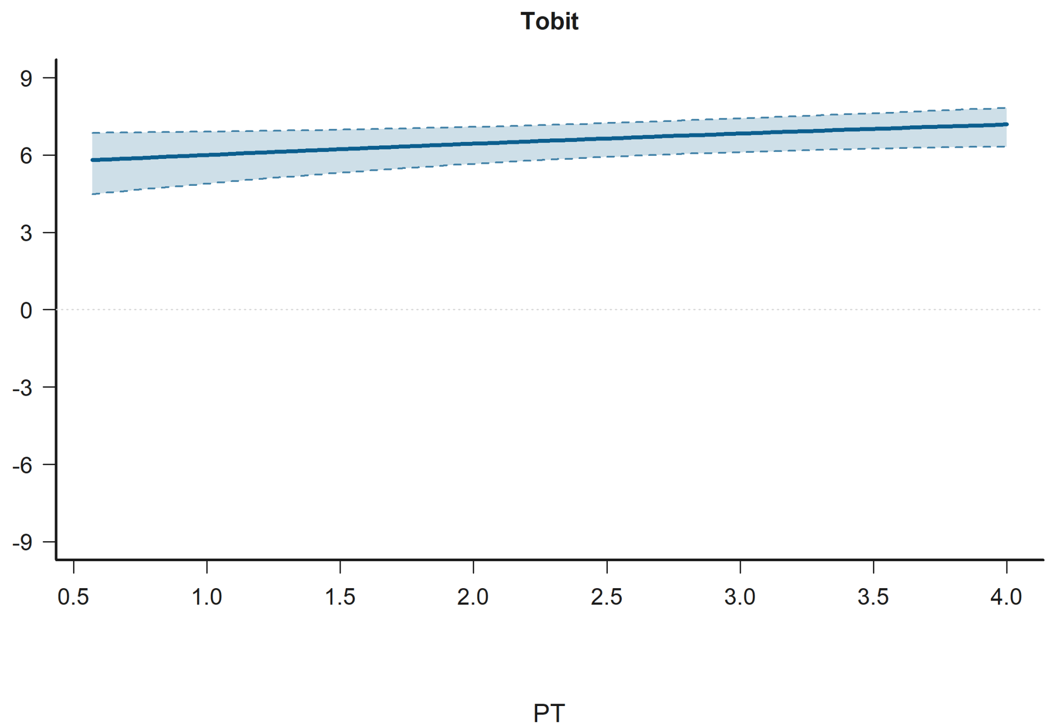

Figure 25: Effect Plot for Empathy: Perspective taking in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) is statistically significant in:

- H2: Negotiator-side relational structure Model A (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)

Figure 26 shows that predicted judgment is lower for JN Out, CN In than for Ref: JN In, CN In.

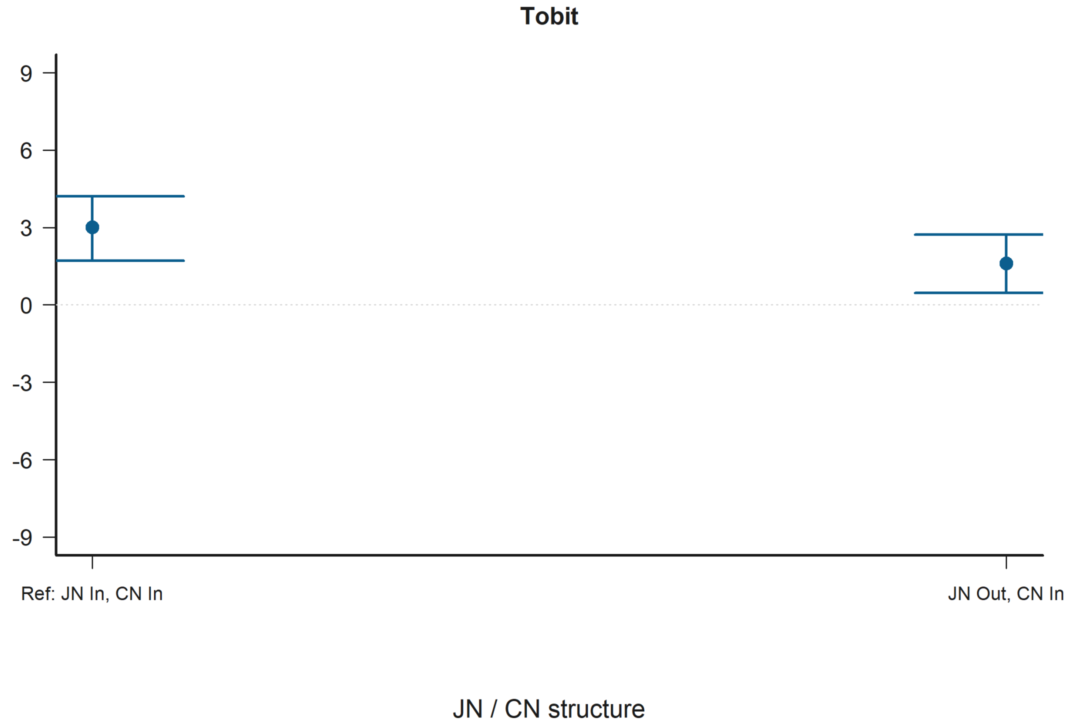

Figure 26: Grouped Prediction Plot for Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) in the judgments_bystander. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

judgments_victim Negotiator accepted harmful deal is statistically significant in:

- H1: Empathy under relational controls Model B (Tobit)
- H1: Empathy under relational controls Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 27 shows that predicted judgment is lower for Acc than for Rej.

Figure 27: Grouped Prediction Plot for Negotiator accepted harmful deal in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Perspective taking is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 28 shows that across the observed range, higher Empathy: Perspective taking corresponds to higher predicted judgment.

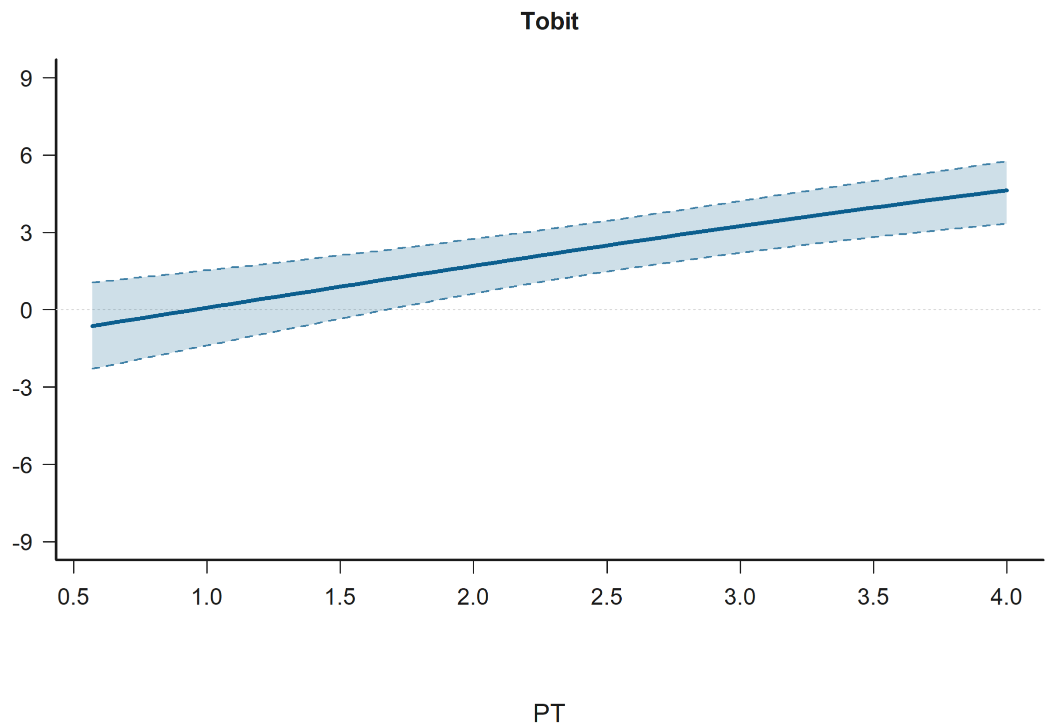

Figure 28: Effect Plot for Empathy: Perspective taking in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Empathic concern is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 29 shows that across the observed range, higher Empathy: Empathic concern corresponds to lower predicted judgment.

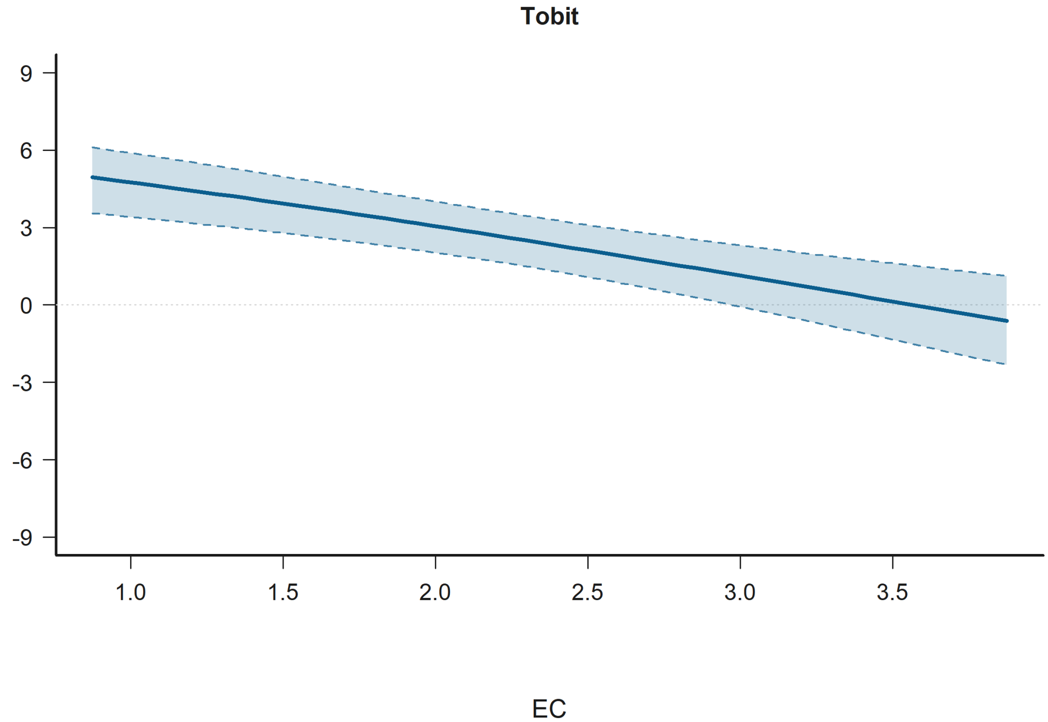

Figure 29: Effect Plot for Empathy: Empathic concern in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy: Personal distress is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 30 shows that across the observed range, higher Empathy: Personal distress corresponds to higher predicted judgment.

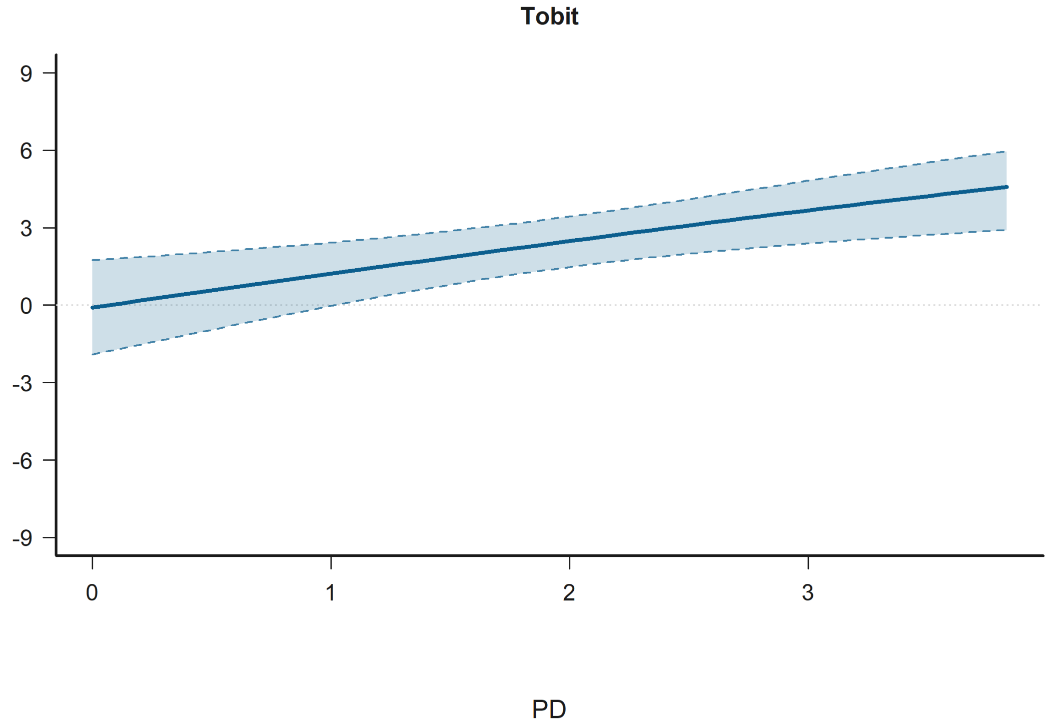

Figure 30: Effect Plot for Empathy: Personal distress in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Engineering participant (ref = humanities) is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)
- H1: Empathy under relational controls Model A (Tobit)
- H3: Empathy x judged-status moderation Model B (Tobit)
- H1: Empathy under relational controls Model B (Tobit)
- H2: Negotiator-side relational structure Model A (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)

Figure 31 shows that predicted judgment is higher for Eng part. than for Hum part..

Figure 31: Grouped Prediction Plot for Engineering participant (ref = humanities) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Judged negotiator outgroup (ref = ingroup) is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)
- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 32 shows that predicted judgment is lower for JN Out than for JN In.

Figure 32: Grouped Prediction Plot for Judged negotiator outgroup (ref = ingroup) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy composite (average) is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 33 shows that across the observed range, higher Empathy composite (average) corresponds to higher predicted judgment.

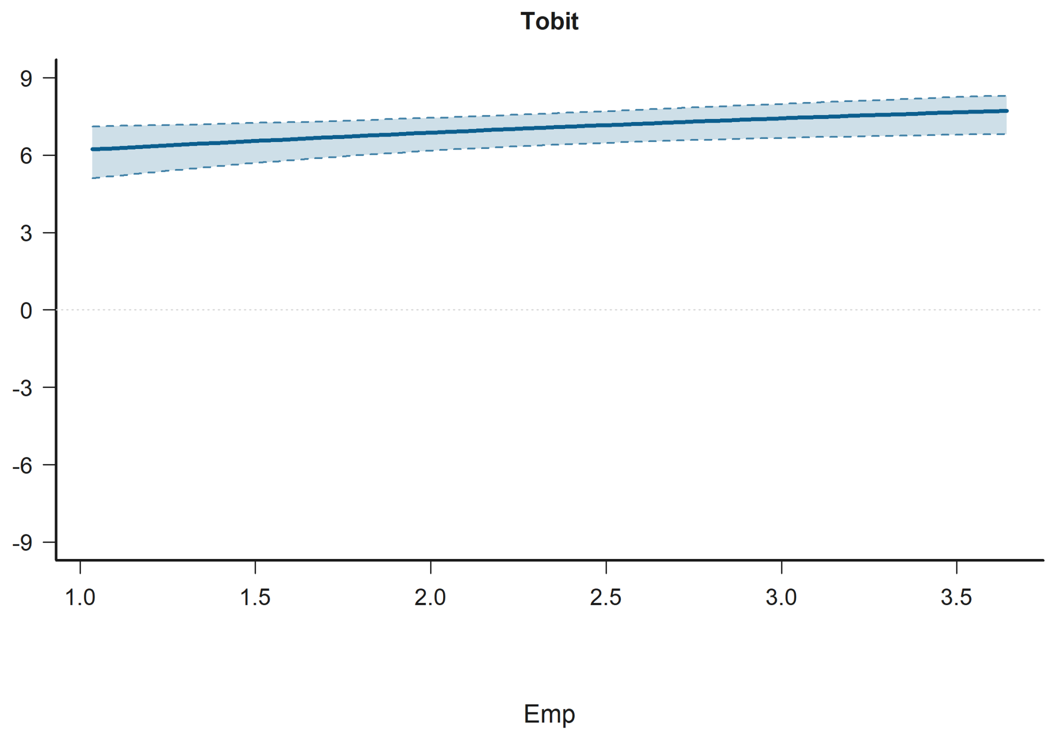

Figure 33: Effect Plot for Empathy composite (average) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 34 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is JN In.

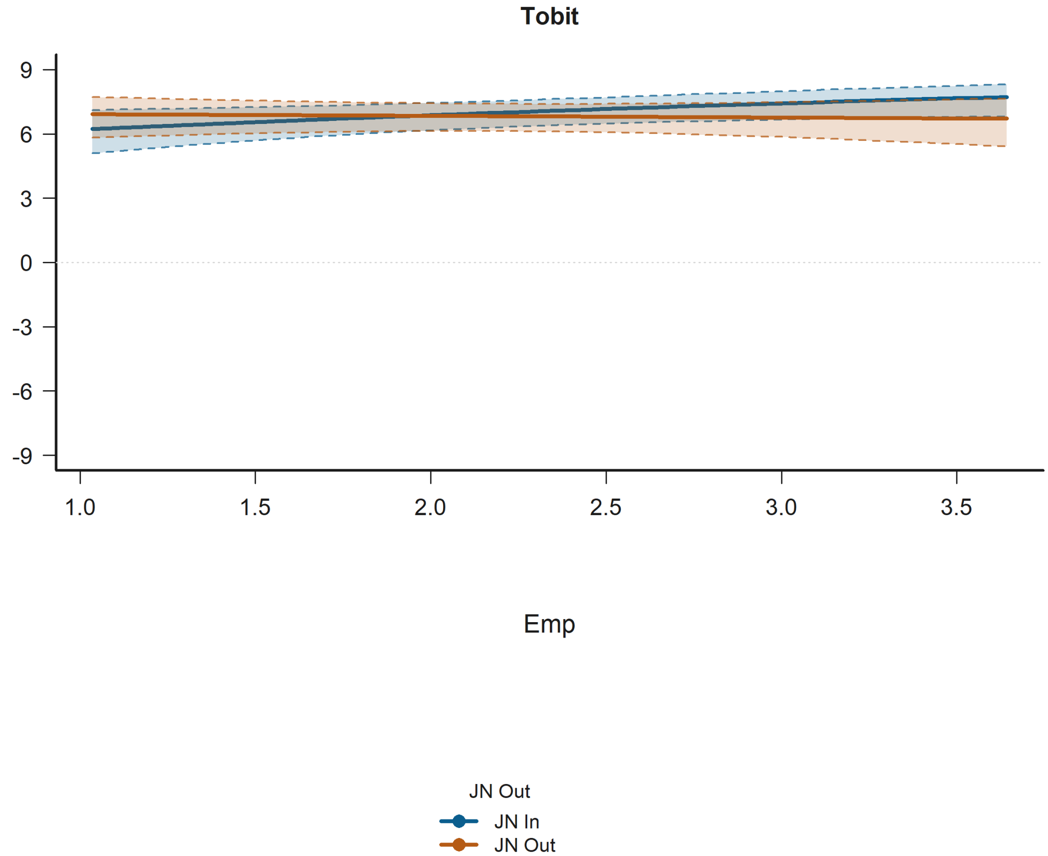

Figure 34: Interaction Plot for Empathy x judged negotiator outgroup in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathy x judged negotiator control label hidden is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 35 shows that the predicted relationship rises most sharply for Empathy composite (average) when the condition is JN In.

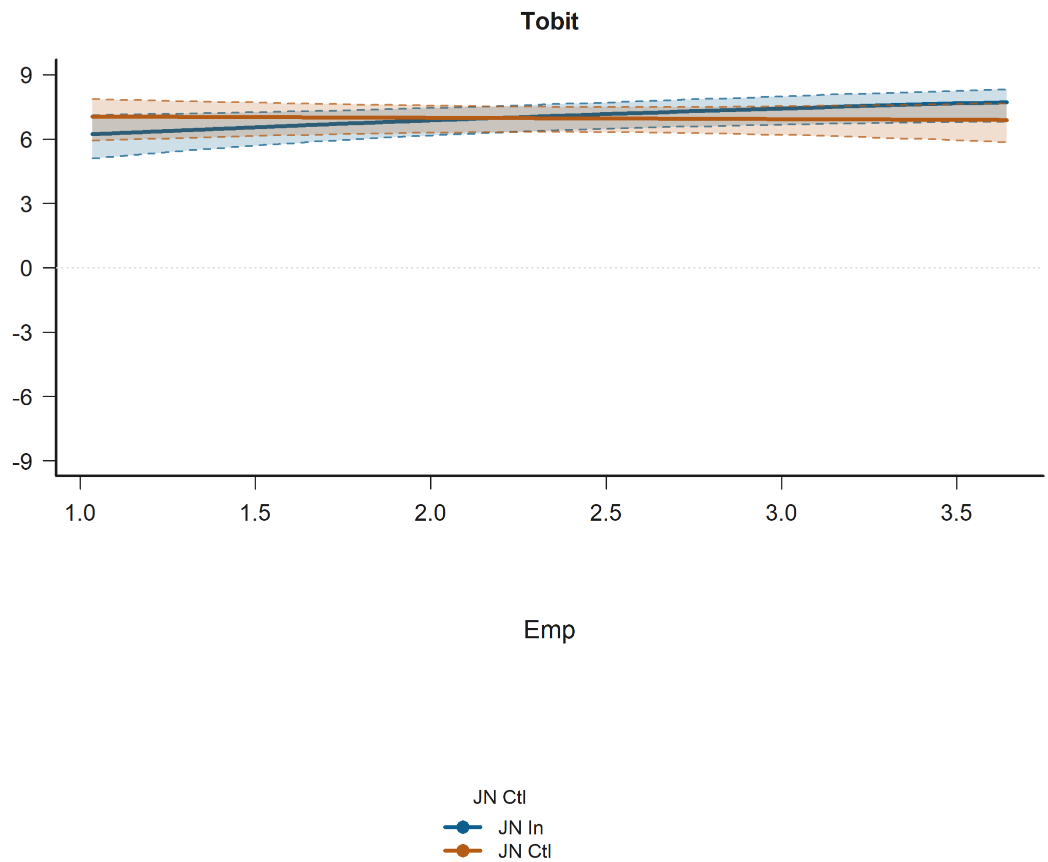

Figure 35: Interaction Plot for Empathy x judged negotiator control label hidden in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Empathic concern x judged negotiator outgroup is statistically significant in:

- H3: Empathy x judged-status moderation Model B (Tobit)

Figure 36 shows that the predicted relationship falls most sharply for Empathy: Empathic concern when the condition is JN Out.

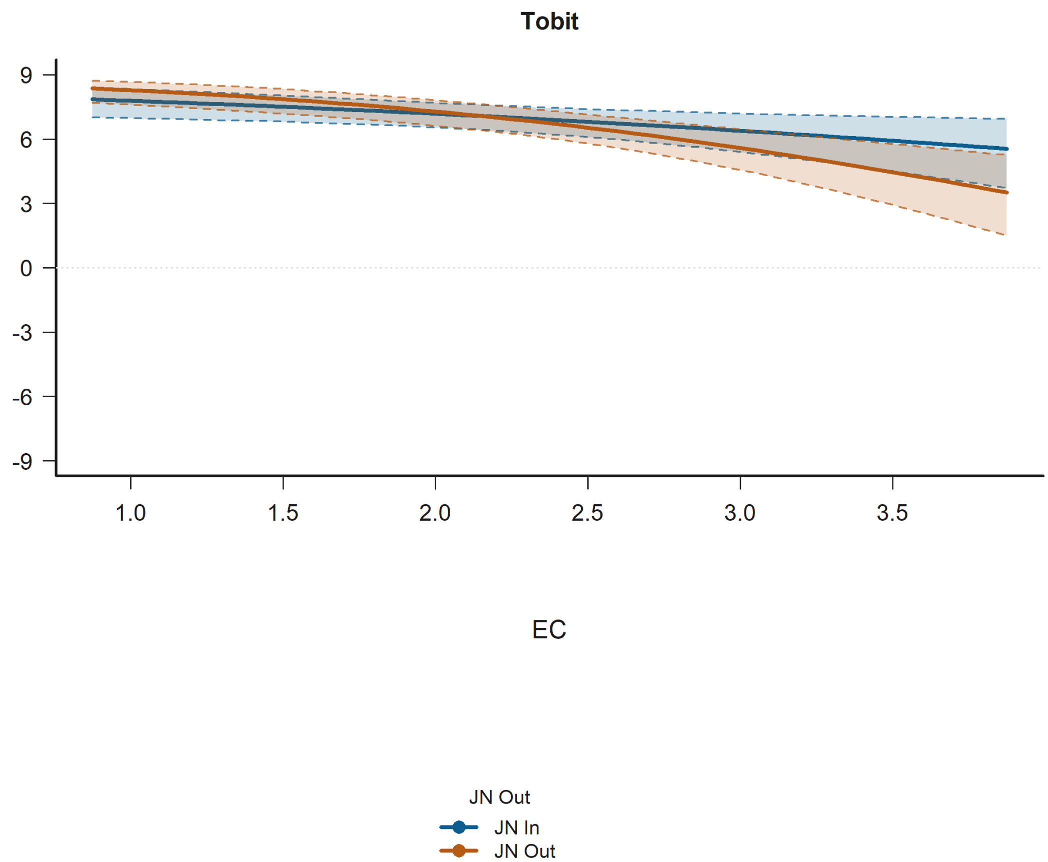

Figure 36: Interaction Plot for Empathic concern x judged negotiator outgroup in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Judged negotiator control label hidden (ref = ingroup) is statistically significant in:

- H3: Empathy x judged-status moderation Model A (Tobit)

Figure 37 shows that predicted judgment is lower for JN Ctl than for JN In.

Figure 37: Grouped Prediction Plot for Judged negotiator control label hidden (ref = ingroup) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Socioeconomic status is statistically significant in:

- H2: Negotiator-side relational structure Model A (Tobit)
- H2: Negotiator-side relational structure Model B (Tobit)

Figure 38 shows that predicted judgment is higher for Socioeconomic status 5 than for Socioeconomic status 0.

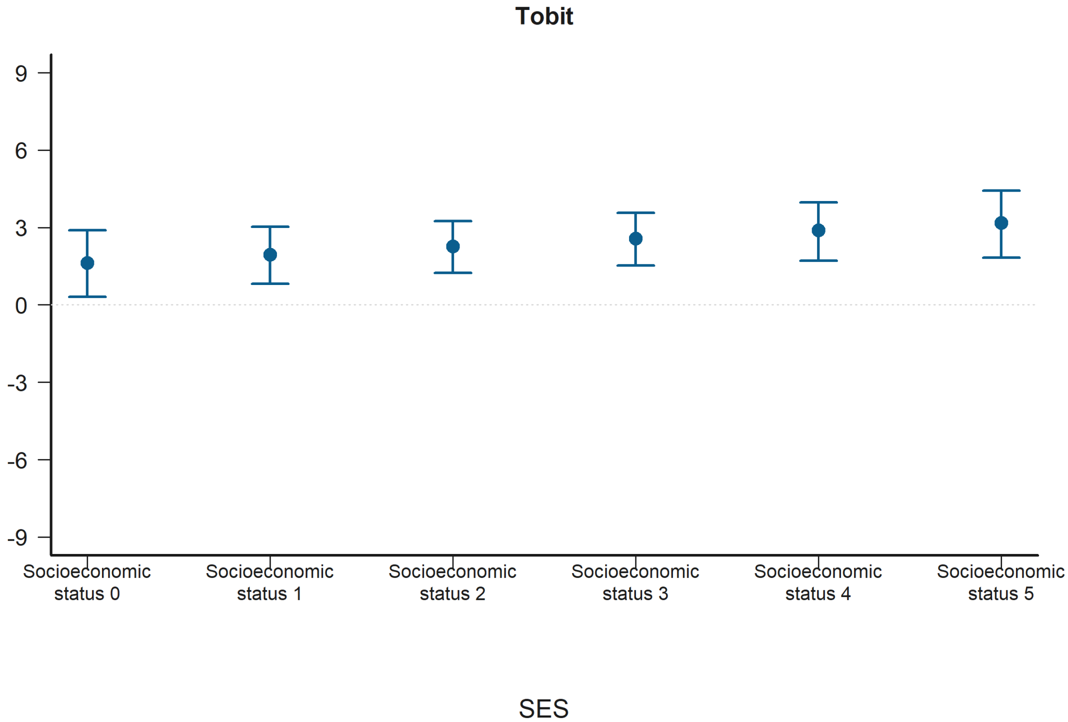

Figure 38: Grouped Prediction Plot for Socioeconomic status in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) is statistically significant in:

- H2: Negotiator-side relational structure Model B (Tobit)

Figure 39 shows that predicted judgment is lower for JN Ctl, CN In than for Ref: JN In, CN In.

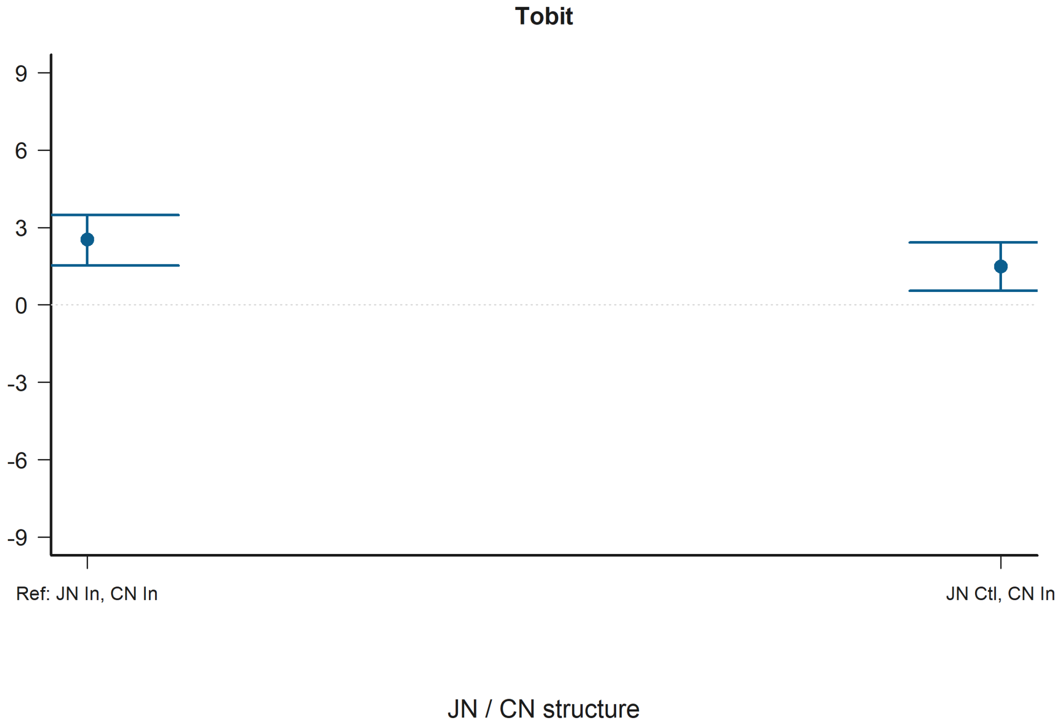

Figure 39: Grouped Prediction Plot for Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) in the judgments_victim. The panels show predicted judgments on the observed -9 to 9 scale with 95% confidence intervals. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

8.4 Integrated Hypothesis Conclusions

The following summary restates each original hypothesis and indicates whether the available Tobit estimates support it in the current data.

- H1. Original hypothesis: Within the victim and bystander subsets, higher empathy predicts lower moral-judgment scores for harmful decisions after conditioning on judged-negotiator status, counterpart status, decision outcome, observer-side victim alignment when applicable, and participant controls. Victim subset: Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A does not support the hypothesis; Empathy composite (average) is positive but not statistically significant ($p = 0.513$). Model B supports the hypothesis through Empathy: Empathic concern with a negative association ($p < 0.001$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Engineering participant (ref = humanities) with a positive association ($p = 0.005$). Bystander subset: Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A does not support the hypothesis; Empathy composite (average) is negative but not statistically significant ($p = 0.952$). Model B supports the hypothesis through Empathy: Empathic concern with a negative association ($p = 0.045$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Engineering participant (ref = humanities) with a positive association ($p = 0.015$).

- H2. Original hypothesis: Moral-judgment severity should vary with the judgment-level ingroup/outgroup/control structure of the judged and counterpart negotiators. In the bystander subset, that negotiator-side structure should further depend on whether the player and the victim share faculty or not. Victim subset: Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; none of the negotiator-side ingroup/outgroup/control structure dummies in the victim subset are statistically significant, and the closest signal is Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) with a negative association ($p = 0.121$). Model B does not support the hypothesis; none of the negotiator-side ingroup/outgroup/control structure dummies in the victim subset are statistically significant, and the closest signal is Negotiator-side structure: judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) with a negative association ($p = 0.087$). Additional statistically significant signals include Empathy: Perspective taking with a positive association ($p < 0.001$) and Empathy: Empathic concern with a negative association ($p < 0.001$). Bystander subset: Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; none of the negotiator-side structure, player-victim outgroup term, and their interaction in the bystander subset are statistically significant, and the closest signal is Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) with a negative association ($p = 0.065$). Model B does not support the hypothesis; none of the negotiator-side structure, player-victim outgroup term, and their interaction in the bystander subset are statistically significant, and the closest signal is Negotiator-side structure: judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) with a negative association ($p = 0.087$). Additional statistically significant signals include Age with a positive association ($p = 0.001$) and Empathy: Empathic concern with a negative association ($p = 0.001$).

- H3. Original hypothesis: The empathy effect may vary according to whether the judged negotiator is ingroup, outgroup, or control, while decision outcome and the additional relational controls remain explicitly modeled. Victim subset: Tobit conclusion: the evidence is mixed but offers partial support for the hypothesis. Model A supports the hypothesis through Empathy x judged negotiator outgroup with a negative association ($p = 0.028$) and Empathy x judged negotiator control label hidden with a negative association ($p = 0.043$). Model B does not support the hypothesis; none of the empathy-dimension x judged-negotiator relational-status interactions are statistically significant, and the closest signal is Empathic concern x judged negotiator outgroup with a negative association ($p = 0.053$). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Empathy: Perspective taking with a positive association ($p < 0.001$). Bystander subset: Tobit conclusion: the available models support the hypothesis. Model A supports the hypothesis through Empathy x judged negotiator control label hidden with a positive association ($p = 0.015$). Model B supports the hypothesis through Personal distress x judged negotiator outgroup with a positive association (p = 0.021). Additional statistically significant signals include Negotiator accepted harmful deal with a negative association ($p < 0.001$) and Judged negotiator control label hidden (ref = ingroup) with a negative association (p = 0.010).

9 Hypothesis Validation and Results

Detailed coefficient tables for each Tobit model, including predictor estimates, standard errors, p-values with conventional significance symbols, and estimator diagnostics for the victim and bystander subsets, are provided below.

9.1 H1: Empathy Effect

This section evaluates H1 separately in the victim and bystander subsets. Model A uses the overall empathy composite. Model B replaces that composite with the four IRI subscales: fantasy, empathic concern, perspective taking, and personal distress. Both formulas retain judged-negotiator status, counterpart status, decision outcome, and participant controls for sex, age, and economic status, but only the bystander subset includes the observer-side victim-alignment predictor because it is not meaningful inside the victim-only subset.

9.1.1 Model A: Composite Empathy

Model A subset-specific formulas.

Victim: judgement
textasciitilde iri_total + judged_outgroup + judged_control + counterpart_outgroup + counterpart_control + decision_accept + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Bystander: judgement
textasciitilde iri_total + judged_outgroup + judged_control + counterpart_outgroup + counterpart_control + observer_victim_outgroup + decision_accept + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Victim subset

Table 6: H1 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Victim subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	7.117	2.323	0.002**
Emp	0.364	0.557	0.513
JN Out	-0.207	0.394	0.599
JN Ctl	0.335	0.398	0.400
CN Out	-0.368	0.373	0.323
CN Ctl	-0.379	0.402	0.346
Acc	-15.014	0.867	<0.001***
Eng part.	1.384	0.493	0.005**
Man	-0.099	0.512	0.847
Age	0.044	0.100	0.659
SES	0.261	0.248	0.293
Slot 2	0.106	0.301	0.723
Log(scale)	2.003	0.049	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 7.117 (p=0.002). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.364. The p-value of 0.513 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.207. The p-value of 0.599 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.335. The p-value of 0.400 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.368. The p-value of 0.323 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.379. The p-value of 0.346 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -15.014. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.384. The p-value of 0.005 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.099. The p-value of 0.847 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.044. The p-value of 0.659 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.261. The p-value of 0.293 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 2.003, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.935$, $p = 3.301e-32$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 7: H1 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Bystander subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	5.347	2.309	0.021*
Emp	-0.031	0.513	0.952
JN Out	-0.349	0.341	0.306
JN Ctl	-0.077	0.350	0.826
CN Out	-0.261	0.357	0.465
CN Ctl	0.463	0.351	0.188
V Out (Obs)	-0.149	0.331	0.654
Acc	-13.139	0.794	<0.001***
Eng part.	1.173	0.482	0.015*
Man	-0.157	0.500	0.753
Age	0.155	0.097	0.112
SES	0.099	0.223	0.657
Slot 2	0.376	0.285	0.186
Log(scale)	1.926	0.048	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 5.347 (p=0.021). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.031. The p-value of 0.952 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.349. The p-value of 0.306 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.077. The p-value of 0.826 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.261. The p-value of 0.465 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.463. The p-value of 0.188 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.149. The p-value of 0.654 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.139. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.173. The p-value of 0.015 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.157. The p-value of 0.753 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.155. The p-value of 0.112 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.099. The p-value of 0.657 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.926, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.945, p = 6.061e-30). The primary Tobit model still uses participant-clustered standard errors by id,.

9.1.2 Model B: Separated Empathy Constructs

Model B subset-specific formulas.

Victim: judgement
textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + judged_outgroup + judged_control + counterpart_outgroup + counterpart_control + decision_accept + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Bystander: judgement
textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + judged_outgroup + judged_control + coun-

terpart_outgroup + counterpart_control + observer_victim_outgroup + decision_accept + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Victim subset

Table 8: H1 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Victim subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	6.802	2.293	0.003**
FS	-0.640	0.462	0.166
EC	-2.641	0.653	<0.001***
PT	2.016	0.493	<0.001***
PD	1.684	0.592	0.004**
JN Out	-0.129	0.387	0.738
JN Ctl	0.204	0.395	0.605
CN Out	-0.290	0.360	0.420
CN Ctl	-0.500	0.390	0.200
Acc	-14.876	0.854	<0.001***
Eng part.	1.196	0.478	0.012*
Man	-0.107	0.527	0.839
Age	0.051	0.094	0.583
SES	0.209	0.234	0.372
Slot 2	0.079	0.299	0.793
Log(scale)	1.986	0.048	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 6.802 (p=0.003). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.640. The p-value of 0.166 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.641. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 2.016. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.684. The p-value of 0.004 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.129. The p-value of 0.738 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.204. The p-value of 0.605 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.290. The p-value of 0.420 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.500. The p-value of 0.200 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -14.876. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.196. The p-value of 0.012 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.107. The p-value of 0.839 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.051. The p-value of 0.583 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.209. The p-value of 0.372 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.986, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.944$, $p = 4.180\text{e-}30$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 9: H1 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Bystander subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	4.981	2.328	0.032*
FS	-0.511	0.463	0.269
EC	-1.435	0.716	0.045*
PT	0.871	0.440	0.048*
PD	1.110	0.563	0.049*
JN Out	-0.270	0.339	0.425
JN Ctl	-0.024	0.349	0.945
CN Out	-0.180	0.350	0.607
CN Ctl	0.508	0.351	0.148
V Out (Obs)	-0.238	0.330	0.471
Acc	-13.061	0.792	<0.001***
Eng part.	1.032	0.468	0.027*
Man	-0.074	0.508	0.885
Age	0.162	0.099	0.102
SES	0.081	0.221	0.712
Slot 2	0.381	0.283	0.178
Log(scale)	1.919	0.047	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 4.981 (p=0.032). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.511. The p-value of 0.269 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.435. The p-value of 0.045 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.871. The p-value of 0.048 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.110. The p-value of 0.049 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.270. The p-value of 0.425 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.024. The p-value of 0.945 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.180. The p-value of 0.607 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.508. The p-value of 0.148 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.238. The p-value of 0.471 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.061. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.032. The p-value of 0.027 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.074. The p-value of 0.885 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.162. The p-value of 0.102 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.081. The p-value of 0.712 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.919, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.950, p = 7.200e-29). The primary Tobit model still uses participant-clustered standard errors by id,.

9.2 H2: Negotiator-Side Relational Structure

This section estimates H2 separately in the victim and bystander subsets. The dependent variable remains negotiator-specific judgment severity, so each participant contributes two judgment rows per scenario, one for each negotiator. In the victim subset, H2 uses the joint judged-plus-counterpart negotiator structure relative to the victim-player, with judged ingroup and counterpart ingroup as the reference structure. In the bystander subset, the same negotiator-side structure is defined relative to the observing player's faculty, and the model additionally includes the player-victim outgroup term and its interaction with the negotiator-side structure dummies. Victim-subset equation: $y^*_{isj,Victim} = \beta_0 + \beta_1 \text{Empathy}_i + \boldsymbol{\gamma}'\mathbf{S}^{(V)}_{isj} + \boldsymbol{\delta}'\mathbf{Z}_i + \epsilon_{isj}$, where $\mathbf{S}^{(V)}_{isj}$ indexes the judged-plus-counterpart structure dummies and $\mathbf{Z}_i$ collects participant controls. Bystander-subset equation: $y^*_{isj,Obs} = \beta_0 + \beta_1 \text{Empathy}_i + \boldsymbol{\gamma}'\mathbf{S}^{(O)}_{isj} + \eta V_{is} + \boldsymbol{\theta}'(\mathbf{S}^{(O)}_{isj} \times V_{is}) + \boldsymbol{\delta}'\mathbf{Z}_i + \epsilon_{isj}$, where V_{is} is the player-victim outgroup indicator.

9.2.1 Model A: Composite Empathy Control

Model A subset-specific formulas.

Victim: judgement
textasciitilde iri_total + h2_negstruct_j_out_c_in + h2_negstruct_j_cont_c_in + h2_negstruct_j_in_c_out + h2_negstruct_j_out_c_out + h2_negstruct_j_cont_c_out + h2_negstruct_j_in_c_cont + h2_negstruct_j_out_c_cont + h2_negstruct_j_cont_c_cont + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Bystander: judgement
textasciitilde iri_total + h2_negstruct_j_out_c_in + h2_negstruct_j_cont_c_in + h2_negstruct_j_in_c_out + h2_negstruct_j_out_c_out + h2_negstruct_j_cont_c_out + h2_negstruct_j_in_c_cont + h2_negstruct_j_out_c_cont + h2_negstruct_j_cont_c_cont + player_victim_outgroup + player_victim_outgroup:h2_negstruct_j_out_c_in + player_victim_outgroup:h2_negstruct_j_cont_c_in + player_victim_outgroup:h2_negstruct_j_in_c_out + player_victim_outgroup:h2_negstruct_j_out_c_out + player_victim_outgroup:h2_negstruct_j_cont_c_out + player_victim_outgroup:h2_negstruct_j_in_c_cont + player_victim_outgroup:h2_negstruct_j_out_c_cont + player_victim_outgroup:h2_negstruct_j_cont_c_cont + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Victim subset

Table 10: H2 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Victim subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	-0.261	3.093	0.933
Emp	0.542	0.730	0.458
JN Out, CN In	-0.269	1.037	0.796
JN Ctl, CN In	-1.732	1.118	0.121
JN In, CN Out	-0.288	1.009	0.775
JN Out, CN Out	-1.155	1.182	0.328
JN Ctl, CN Out	-0.933	1.073	0.384
JN In, CN Ctl	-0.691	1.129	0.541
JN Out, CN Ctl	-1.081	1.141	0.344
JN Ctl, CN Ctl	-0.394	1.047	0.707
Eng part.	1.572	0.675	0.020*
Man	0.187	0.720	0.795
Age	0.015	0.132	0.909
SES	0.587	0.324	0.070+
Slot 2	-0.172	0.438	0.695
Log(scale)	2.449	0.055	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -0.261 (p=0.933). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 0.542. The p-value of 0.458 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_in (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref

= judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.269. The p-value of 0.796 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_in (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.732. The p-value of 0.121 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.288. The p-value of 0.775 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.155. The p-value of 0.328 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.933. The p-value of 0.384 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.691. The p-value of 0.541 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.081. The p-value of 0.344 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.394. The p-value of 0.707 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.572. The p-value of 0.020 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.187. The p-value of 0.795 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.015. The p-value of 0.909 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.587. The p-value of 0.070 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 2.449, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.908$, $p = 6.503e\text{-}37$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 11: H2 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Bystander subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	-3.294	2.679	0.219
Emp	-0.708	0.610	0.246
JN Out, CN In	-2.363	1.281	0.065+
JN Ctl, CN In	1.111	1.191	0.351
JN In, CN Out	-1.201	1.329	0.366
JN Out, CN Out	-0.959	1.485	0.518
JN Ctl, CN Out	-0.400	1.407	0.776
JN In, CN Ctl	-0.336	1.287	0.794
JN Out, CN Ctl	-0.382	1.389	0.783
JN Ctl, CN Ctl	0.176	1.403	0.900
V Out	-1.623	1.467	0.269
Eng part.	1.125	0.569	0.048*
Man	0.694	0.614	0.258
Age	0.325	0.101	0.001**
SES	0.379	0.278	0.173
Slot 2	0.714	0.434	0.100+
JN Out, CN In x V Out	2.349	2.098	0.263
JN Ctl, CN In x V Out	0.372	1.856	0.841
JN In, CN Out x V Out	2.694	1.990	0.176
JN Out, CN Out x V Out	3.543	2.214	0.110
JN Ctl, CN Out x V Out	-1.366	1.954	0.485
JN In, CN Ctl x V Out	-0.479	1.902	0.801
JN Out, CN Ctl x V Out	-0.184	1.912	0.923
JN Ctl, CN Ctl x V Out	0.727	1.930	0.706
Log(scale)	2.318	0.055	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -3.294 (p=0.219). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.708. The p-value of 0.246 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_in (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -2.363. The p-value of 0.065 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_in (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 1.111. The p-value of 0.351 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.201. The p-value of 0.366 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.959. The p-value of 0.518 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.400. The p-value of 0.776 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.336. The p-value of 0.794 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.382. The p-value of 0.783 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 0.176. The p-value of 0.900 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term player_victim_outgroup (representing whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -1.623. The p-value of 0.269 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.125. The p-value of 0.048 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.694. The p-value of 0.258 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.325. The p-value of 0.001 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.379. The p-value of 0.173 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_in:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 2.349. The p-value of 0.263 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_in:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 0.372. The p-value of 0.841 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_out:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 2.694. The p-value of 0.176 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_out:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 3.543. The p-value of 0.110 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_out:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -1.366. The p-value of 0.485 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_cont:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -0.479. The p-value of 0.801 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_cont:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -0.184. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_cont:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 0.727. The p-value of 0.706 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 2.318, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.931, p = 5.366e-33). The primary Tobit model still uses participant-clustered standard errors by id,.

9.2.2 Model B: Separated Construct Controls

Model B subset-specific formulas.

Victim: judgement
textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + h2_negstruct_j_out_c_in + h2_negstruct_j_cont_c_in
+ h2_negstruct_j_in_c_out + h2_negstruct_j_out_c_out + h2_negstruct_j_cont_c_out
+ h2_negstruct_j_in_c_cont + h2_negstruct_j_out_c_cont + h2_negstruct_j_cont_c_cont
+ participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Bystander: judgement
textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + h2_negstruct_j_out_c_in + h2_negstruct_j_cont_c_in
+ h2_negstruct_j_in_c_out + h2_negstruct_j_out_c_out + h2_negstruct_j_cont_c_out
+ h2_negstruct_j_in_c_cont + h2_negstruct_j_out_c_cont + h2_negstruct_j_cont_c_cont
+ player_victim_outgroup + player_victim_outgroup:h2_negstruct_j_out_c_in + player_victim_outgroup:h2_ne
+ player_victim_outgroup:h2_negstruct_j_in_c_out + player_victim_outgroup:h2_negstruct_j_out_c_out
+ player_victim_outgroup:h2_negstruct_j_cont_c_out + player_victim_outgroup:h2_negstruct_j_in_c_cont
+ player_victim_outgroup:h2_negstruct_j_out_c_cont + player_victim_outgroup:h2_negstruct_j_cont_c_cont
+ participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Victim subset

Table 12: H2 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Victim subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	-0.830	2.917	0.776
FS	-1.004	0.627	0.110
EC	-3.535	0.818	<0.001***
PT	2.896	0.644	<0.001***
PD	2.306	0.742	0.002**
JN Out, CN In	-0.087	1.018	0.932
JN Ctl, CN In	-1.895	1.109	0.087+
JN In, CN Out	-0.129	0.982	0.895
JN Out, CN Out	-0.975	1.158	0.400
JN Ctl, CN Out	-0.909	1.064	0.393
JN In, CN Ctl	-0.848	1.105	0.443
JN Out, CN Ctl	-1.059	1.122	0.345
JN Ctl, CN Ctl	-0.737	1.017	0.469
Eng part.	1.321	0.648	0.041*
Man	0.186	0.730	0.799
Age	0.027	0.124	0.827
SES	0.502	0.297	0.091+
Slot 2	-0.191	0.438	0.662
Log(scale)	2.437	0.054	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -0.830 (p=0.776). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.004. The p-value of 0.110 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -3.535. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 2.896. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 2.306. The p-value of 0.002 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term h2_negstruct_j_out_c_in (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.087. The p-value of 0.932 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_in (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.895. The p-value of 0.087 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.129. The p-value of 0.895 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.975. The p-value of 0.400 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.909. The p-value of 0.393 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.848. The p-value of 0.443 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.059. The p-value of 0.345 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.737. The p-value of 0.469 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.321. The p-value of 0.041 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.186. The p-value of 0.799 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.027. The p-value of 0.827 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.502. The p-value of 0.091 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 2.437, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.929$, $p = 2.832e\text{-}33$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 13: H2 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Bystander subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	-3.484	2.685	0.194
FS	-0.689	0.582	0.236
EC	-2.586	0.809	0.001**
PT	0.938	0.537	0.081+
PD	1.664	0.646	0.010**
JN Out, CN In	-2.197	1.284	0.087+
JN Ctl, CN In	1.235	1.197	0.302
JN In, CN Out	-1.035	1.314	0.431
JN Out, CN Out	-0.489	1.450	0.736
JN Ctl, CN Out	-0.147	1.407	0.917
JN In, CN Ctl	-0.207	1.287	0.872
JN Out, CN Ctl	-0.134	1.388	0.923
JN Ctl, CN Ctl	0.174	1.398	0.901
V Out	-1.783	1.453	0.220
Eng part.	0.959	0.548	0.080+
Man	0.823	0.614	0.180
Age	0.333	0.102	0.001**
SES	0.338	0.267	0.207
Slot 2	0.716	0.433	0.098+
JN Out, CN In x V Out	2.361	2.106	0.262
JN Ctl, CN In x V Out	0.482	1.853	0.795
JN In, CN Out x V Out	2.697	1.979	0.173
JN Out, CN Out x V Out	3.320	2.213	0.134
JN Ctl, CN Out x V Out	-1.426	1.939	0.462
JN In, CN Ctl x V Out	-0.373	1.897	0.844
JN Out, CN Ctl x V Out	-0.247	1.902	0.897
JN Ctl, CN Ctl x V Out	1.044	1.933	0.589
Log(scale)	2.311	0.055	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -3.484 (p=0.194). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.689. The p-value of 0.236 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -2.586. The p-value of 0.001 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.938. The p-value of 0.081 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.664. The p-value of 0.010 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term h2_negstruct_j_out_c_in (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -2.197. The p-value of 0.087 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_in (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 1.235. The p-value of 0.302 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -1.035. The p-value of 0.431 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.489. The p-value of 0.736 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_out (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.147. The p-value of 0.917 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.207. The p-value of 0.872 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of -0.134. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_cont (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup)) has an estimated β coefficient of 0.174. The p-value of 0.901 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term player_victim_outgroup (representing whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -1.783. The p-value of 0.220 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.959. The p-value of 0.080 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of 0.823. The p-value of 0.180 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.333. The p-value of 0.001 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.338. The p-value of 0.207 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_in:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 2.361. The p-value of 0.262 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_in:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator ingroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 0.482. The p-value of 0.795 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_out:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 2.697. The p-value of 0.173 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_out:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 3.320. The p-value of 0.134 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_out:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator outgroup (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -1.426. The p-value of 0.462 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_in_c_cont:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator ingroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -0.373. The p-value of 0.844 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_out_c_cont:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator outgroup, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of -0.247. The p-value of 0.897 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term h2_negstruct_j_cont_c_cont:player_victim_outgroup (representing the H2 negotiator-side structure contrast comparing judged negotiator control label hidden, counterpart negotiator control label hidden (ref = judged negotiator ingroup, counterpart negotiator ingroup) interacted with whether, in observer-role judgments, the player and victim belonged to different faculties) has an estimated β coefficient of 1.044. The p-value of 0.589 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 2.311, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.937, p = 9.471e-32). The primary Tobit model still uses participant-clustered standard errors by id,.

9.3 H3: Empathy x Judged-Status Moderation

This section tests whether empathy slopes differ across judged-negotiator status categories after retaining decision outcome, judged-status x decision terms, counterpart status, and observer-side victim alignment when that predictor is meaningful. As in H1, the victim and bystander subsets use different formulas only where observer-only predictors would otherwise be structurally fixed.

9.3.1 Model A: Composite Empathy Interaction

Model A subset-specific formulas.

> Victim: judgement
> textasciitilde iri_total + judged_outgroup + judged_control + decision_accept:judged_outgroup + decision_accept:judged_control + iri_total:judged_outgroup + iri_total:judged_control + decision_accept + counterpart_outgroup + counterpart_control + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Bystander: judgement
textasciitilde iri_total + judged_outgroup + judged_control + decision_accept:judged_outgroup + decision_accept:judged_control + iri_total:judged_outgroup + iri_total:judged_control + decision_accept + counterpart_outgroup + counterpart_control + observer_victim_outgroup + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Victim subset

Table 14: H3 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Victim subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	4.689	2.663	0.078+
Emp	1.536	0.690	0.026*
JN Out	3.391	1.798	0.059+
JN Ctl	3.695	2.032	0.069+
Acc	-15.431	0.975	<0.001***
CN Out	-0.376	0.369	0.308
CN Ctl	-0.350	0.402	0.383
Eng part.	1.406	0.493	0.004**
Man	-0.084	0.511	0.870
Age	0.045	0.100	0.650
SES	0.252	0.247	0.307
Slot 2	0.127	0.301	0.673
Acc x JN Out	0.445	0.811	0.583
Acc x JN Ctl	0.751	0.792	0.343
Emp x JN Out	-1.730	0.787	0.028*
Emp x JN Ctl	-1.693	0.837	0.043*
Log(scale)	2.002	0.048	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 4.689 (p=0.078). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of 1.536. The p-value of 0.026 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 3.391. The p-value of 0.059 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 3.695. The p-value of 0.069 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -15.431. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.376. The p-value of 0.308 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.350. The p-value of 0.383 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.406. The p-value of 0.004 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.084. The p-value of 0.870 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.045. The p-value of 0.650 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.252. The p-value of 0.307 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.445. The p-value of 0.583 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.751.

The p-value of 0.343 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:judged_outgroup (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -1.730. The p-value of 0.028 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the negative coefficient indicates that the slope of empathy on moral judgment is steeper/more negative (more severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The term iri_total:judged_control (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -1.693. The p-value of 0.043 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the negative coefficient indicates that the slope of empathy on moral judgment is steeper/more negative (more severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 2.002, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.935$, $p = 5.037e\text{-}32$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 15: H3 Model A: Composite Empathy Regression Coefficients (Tobit estimator; Bystander subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	6.963	2.575	0.007**
Emp	-0.729	0.647	0.259
JN Out	-1.093	1.497	0.465
JN Ctl	-4.078	1.576	0.010**
Acc	-13.159	0.883	<0.001***
CN Out	-0.286	0.358	0.424
CN Ctl	0.430	0.351	0.221
V Out (Obs)	-0.125	0.330	0.704
Eng part.	1.176	0.483	0.015*
Man	-0.170	0.499	0.734
Age	0.155	0.098	0.113
SES	0.092	0.223	0.681
Slot 2	0.363	0.286	0.204
Acc x JN Out	-0.262	0.722	0.716
Acc x JN Ctl	0.393	0.715	0.583
Emp x JN Out	0.387	0.643	0.547
Emp x JN Ctl	1.706	0.700	0.015*
Log(scale)	1.924	0.048	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 6.963 (p=0.007). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -0.729. The p-value of 0.259 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -1.093. The p-value of 0.465 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -4.078. The p-value of 0.010 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.159. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.286. The p-value of 0.424 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.430. The p-value of 0.221 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.125. The p-value of 0.704 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.176. The p-value of 0.015 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.170. The p-value of 0.734 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.155. The p-value of 0.113 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.092. The p-value of 0.681 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.262. The p-value of 0.716 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.393. The p-value of 0.583 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:judged_outgroup (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of

0.387. The p-value of 0.547 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:judged_control (representing the average composite of empathetic propensity across the participant interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 1.706. The p-value of 0.015 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the positive coefficient indicates that the slope of empathy on moral judgment is more positive (less severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.924, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.946, p = 8.699e-30). The primary Tobit model still uses participant-clustered standard errors by id,.

9.3.2 Model B: Separated Constructs Interaction

Model B subset-specific formulas.

Victim: judgement
textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + judged_outgroup + judged_control + decision_accept:judged_outgroup + decision_accept:judged_control + iri_fs:judged_outgroup + iri_ec:judged_outgroup + iri_pt:judged_outgroup + iri_pd:judged_outgroup + iri_fs:judged_control + iri_ec:judged_control + iri_pt:judged_control + iri_pd:judged_control + decision_accept + counterpart_outgroup + counterpart_control + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Bystander: judgement
textasciitilde iri_fs + iri_ec + iri_pt + iri_pd + judged_outgroup + judged_control + decision_accept:judged_outgroup + decision_accept:judged_control + iri_fs:judged_outgroup + iri_ec:judged_outgroup + iri_pt:judged_outgroup + iri_pd:judged_outgroup + iri_fs:judged_control + iri_ec:judged_control + iri_pt:judged_control + iri_pd:judged_control + decision_accept + counterpart_outgroup + counterpart_control + observer_victim_outgroup + participant_engineering + sex_man + age + economic_status + factor(negotiator_slot)

Victim subset

Table 16: H3 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Victim subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	4.130	2.686	0.124
FS	-0.416	0.659	0.528
EC	-1.915	0.793	0.016*
PT	2.487	0.722	<0.001***
PD	1.413	0.700	0.044*
JN Out	4.009	1.773	0.024*
JN Ctl	3.757	2.342	0.109
Acc	-15.207	0.950	<0.001***
CN Out	-0.279	0.359	0.437
CN Ctl	-0.473	0.389	0.224
Eng part.	1.220	0.478	0.011*
Man	-0.131	0.527	0.804
Age	0.057	0.095	0.549
SES	0.201	0.234	0.391
Slot 2	0.091	0.298	0.761
Acc x JN Out	0.243	0.779	0.755
Acc x JN Ctl	0.662	0.767	0.388
FS x JN Out	-0.715	0.732	0.328
EC x JN Out	-1.887	0.975	0.053+
PT x JN Out	-0.453	0.791	0.567
PD x JN Out	1.299	0.837	0.121
FS x JN Ctl	-0.007	0.852	0.993
EC x JN Ctl	-0.273	1.135	0.810
PT x JN Ctl	-0.890	0.920	0.334
PD x JN Ctl	-0.496	0.820	0.545
Log(scale)	1.983	0.047	<0.001***

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 4.130 (p=0.124). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.416. The p-value of 0.528 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.915. The p-value of 0.016 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 2.487. The p-value of 0.001 indicates highly statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 1.413. The p-value of 0.044 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 4.009. The p-value of 0.024 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 3.757. The p-value of 0.109 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -15.207. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.279. The p-value of 0.437 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.473. The p-value of 0.224 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.220. The p-value of 0.011 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.131. The p-value of 0.804 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.057. The p-value of 0.549 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.201. The p-value of 0.391 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 0.243. The p-value of 0.755 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.662. The p-value of 0.388 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:judged_outgroup (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.715. The p-value of 0.328 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:judged_outgroup (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -1.887. The p-value of 0.053 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. Because this is a continuous-by-discrete interaction, the negative coefficient indicates that the slope of empathy on moral judgment is steeper/more negative (more severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The term iri_pt:judged_outgroup (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.453. The p-value of 0.567 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:judged_outgroup (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 1.299. The p-value of 0.121 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:judged_control (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.007. The p-value of 0.993 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:judged_control (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.273. The p-value of 0.810 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:judged_control (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.890. The p-value of 0.334 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:judged_control (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -0.496. The p-value of 0.545 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.983, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption ($W = 0.945$, $p = 6.155e-30$). The primary Tobit model still uses participant-clustered standard errors by id,.

Bystander subset

Table 17: H3 Model B: Separated Constructs Regression Coefficients (Tobit estimator; Bystander subset: 2,570 observations from 257 participants).

Predictor	Estimate	Std. Error	p-value
Intercept	6.340	2.562	0.013*
FS	-0.648	0.578	0.262
EC	-1.261	0.813	0.121
PT	0.783	0.555	0.158
PD	0.498	0.675	0.461
JN Out	-0.505	1.736	0.771
JN Ctl	-4.124	1.765	0.019*
Acc	-13.048	0.880	<0.001***
CN Out	-0.163	0.350	0.641
CN Ctl	0.481	0.349	0.167
V Out (Obs)	-0.212	0.328	0.519
Eng part.	1.035	0.466	0.026*
Man	-0.094	0.507	0.853
Age	0.163	0.098	0.097+
SES	0.072	0.218	0.742
Slot 2	0.376	0.287	0.191
Acc x JN Out	-0.411	0.708	0.561
Acc x JN Ctl	0.379	0.715	0.596
FS x JN Out	-0.039	0.703	0.956
EC x JN Out	-0.707	0.839	0.399
PT x JN Out	-0.436	0.707	0.537
PD x JN Out	1.576	0.682	0.021*
FS x JN Ctl	0.316	0.714	0.658
EC x JN Ctl	0.288	0.787	0.714
PT x JN Ctl	0.755	0.670	0.260
PD x JN Ctl	0.342	0.662	0.606

Tobit estimator

Note. Abbrev.: Emp = empathy composite; FS / EC / PT / PD = IRI subscales; JN = judged negotiator; CN = counterpart negotiator; V = victim-side player-victim relation in observer models; Vic / Obs = victim / observer subset; In / Out / Ctl = ingroup / outgroup / control label hidden; Acc / Rej = accepted / rejected harmful deal; SES = economic status.

Interpretation:

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at 6.340 (p=0.013). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.648. The p-value of 0.262 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.261. The p-value of 0.121 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.783. The p-value of 0.158 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of 0.498. The p-value of 0.461 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup (representing whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.505. The p-value of 0.771 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control (representing whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of -4.124. The p-value of 0.019 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it) has an estimated β coefficient of -13.048. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term counterpart_outgroup (representing whether the counterpart negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.163. The p-value of 0.641 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term counterpart_control (representing whether the counterpart negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.481. The p-value of 0.167 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term observer_victim_outgroup (representing whether, in observer-role judgments, the victim belonged to a different faculty than the observing participant) has an estimated β coefficient of -0.212. The p-value of 0.519 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 1.035. The p-value of 0.026 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.094. The p-value of 0.853 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.163. The p-value of 0.097 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of 0.072. The p-value of 0.742 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_outgroup:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.411. The p-value of 0.561 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term judged_control:decision_accept (representing whether the judged negotiator accepted the harmful deal instead of rejecting it interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.379. The p-value of 0.596 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:judged_outgroup (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.039. The p-value of 0.956 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:judged_outgroup (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.707. The p-value of 0.399 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:judged_outgroup (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of -0.436. The p-value of 0.537 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:judged_outgroup (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator belonged to a different faculty than the role-relevant reference actor (outgroup versus ingroup)) has an estimated β coefficient of 1.576. The p-value of 0.021 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. Because this is a continuous-by-discrete interaction, the positive coefficient indicates that the slope of empathy on moral judgment is more positive (less severe) for the specified condition compared to the baseline condition. The individual main effects of the components are subsumed by this contextual relationship. The term iri_fs:judged_control (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.316. The p-value of 0.658 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:judged_control (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.288. The p-value of 0.714 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:judged_control (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.755. The p-value of 0.260 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:judged_control (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress) interacted with whether the judged negotiator appeared in the control or unlabeled condition instead of the ingroup condition) has an estimated β coefficient of 0.342. The p-value of 0.606 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.915, capturing the underlying dispersion of latent judgments.

Diagnostics:

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.951, p = 1.754e-28). The primary Tobit model still uses participant-clustered standard errors by id,.

10 Discussion and Limitations

While the dual-estimator strategy strengthens the analysis, several limitations remain. First, both inference strategies assume independence between participants, which may still be violated by broader shared institutional or contextual shocks. Second, we assume a normal distribution for the latent errors ϵ_{ij} in the Tobit branch; departures from normality could affect the consistency of the maximum likelihood estimators. The non-parametric robustness branch reduces dependence on that assumption and adds participant-level cluster bootstrap inference after converged full-sample fits, but bootstrap summaries still carry Monte Carlo error and can be sensitive to convergence problems in difficult specifications. Finally, the use of unstandardized averages for empathy assumes a linear mapping between the psychometric scale and the latent moral preference across both estimators.

11 Conclusion

Based on the combined interval-censored Tobit estimations and the cluster-aware non-parametric robustness workflow, empathy and relational judgment structure have been documented together under Option 2 through negotiator-level relational predictors rather than descriptive case labels.